E-INVOICING GATEWAY API SPECIFICATION

Design, Supply and Installation of VAT Compliance E - Invoicing Solution

Doc. No. v1.11

Submitted to: **Revenue Services Lesotho**
Submitted by: **Norway Registers Development AS**

2026-05-06

Project number:	P1004/P1011
Contract:	Agreement for the Design, Supply, and Installation, Configuration of e-Invoicing Solution, signed on 8 January 2024 (hereinafter – the "Contract")
Task name (s):	MVP: System analysis and design (1/7)

Acceptance Certificate
Between

Revenue Services Lesotho

And

Norway Registers Development AS

18 December 2024

By signing this Acceptance Certificate, we hereby confirm that the deliverables, listed below, are accepted and therefore the corresponding task under the Contract is completed.

Task name as per Contract	Deliverable ID and title submitted	Date delivered
MVP: System analysis and design	1.2.2. System analysis and design	18 December 2024

Revenue Services Lesotho

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1. DOCUMENT SUMMARY

1.1. Purpose

This document describes the E-Invoicing Gateway API to be exposed towards taxpayers Point Of Sales solutions, accounting systems and other e-invoicing devices. It defines exposed methods, input and output parameters, data formats, error codes, communication rules, etc.

1.2. Revision history

Table 1. Document history

Version	Data	Description	Author
0.1	2024-03-19	Initial version	Marius Papšo
1.0	2024-04-11	Version ready for RSL review	Marius Papšo
1.1	2024-05-02	Changed reconciliation period to fiscal period, added Payout document type	Marius Papšo
1.2	2024-08-07	Added changes related to levies	Marius Papšo
1.3	2024-12-18	Testing environment URL updated	Andrius Petrauskas
1.4	2025-01-20	Added line subtype	Marius Papšo
1.5	2025-06-03	Fixed value calculation changes	Marius Papšo
1.6	2025-09-17	Clarifications added regarding tax calculation	Andrius Petrauskas
1.7	2025-10-09	New invoice validation codes for ReceiptCounter and ReceiptGlobalNo added	Andrius Petrauskas
1.8	2025-10-21	New filed in submitReceipt method is added to indicate how tax was calculated	Andrius Petrauskas
1.9	2026-03-12	Tax amount and net amount calculation algorithm change for invoices using per line taxRoundingType calculation method.	Andrius Petrauskas
1.10	2026-03-19	Receipt validation rule RCPT022 updated.	Andrius Petrauskas
1.11	2026-05-06	Receipt validation rule RCPT054 is added (seller and buyer TINs must differ)	Andrius Petrauskas

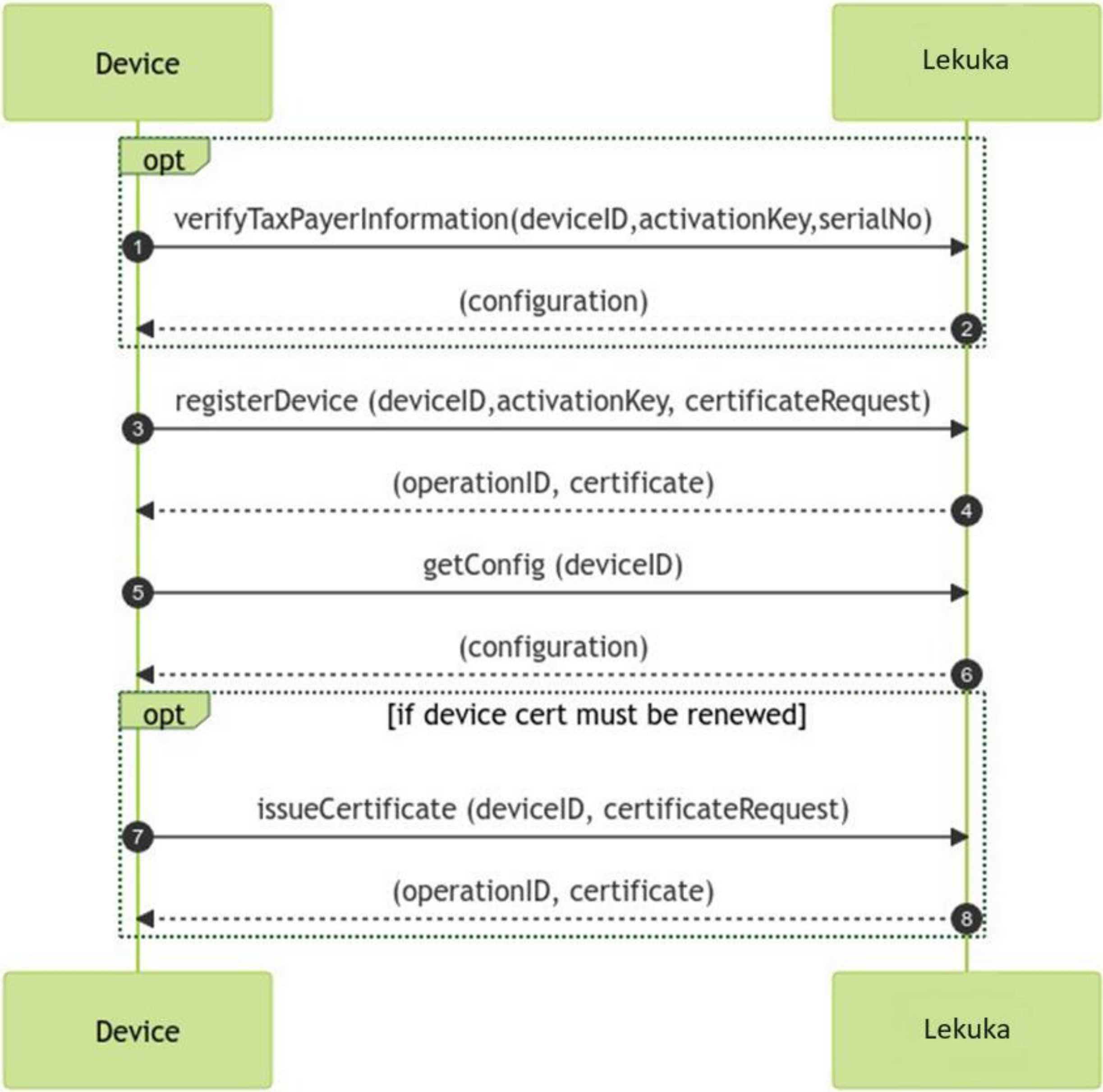
1.3. Definitions

Term	Description
Device	Hardware based or software-based solution which is accounting sales (issues invoices, debit or credit notes) and submits them to Lekuka.
Device signature	Receipt signature signed by device before submission of a receipt to E-Invoicing Gateway API.
E-Invoicing Gateway API	Lekuka system module responsible for invoices, debit, or credit notes acceptance from devices.
E-Invoicing Data Management System (Lekuka)	E-Invoicing Data Management System means system used by the Revenue Services Lesotho to receive, control and monitor User business transactions recorded by point of sales (POS) and accounting systems interfaced to it and to generate various required reports for the purposes of tax revenue administration.
Lekuka signature	Receipt signature signed by Lekuka after submission of a invoices, debit, or credit notes E-Invoicing Gateway API.
QR code	A machine-readable code consisting of an array of black and white squares storing invoices, debit, or credit notes identification information, required to validate a receipt.
QR code data	QR code data is one of few receipt identification fields stored in QR code, which represents invoices, debit, or credit notes device signature.
Receipt	Receipt encompasses invoice, debit, or credit note.
RSL	Revenue Services Lesotho.

2. E-INVOICING GATEWAY USAGE SCENARIOS

2.1. Device registration

The process starts when a taxpayer initially orders POS device using the RSL self-service portal. After completing this process, the taxpayer is provided with device ID and Activation Key. Device registration must be done once before starting to use a new device. After device registration, it needs to get its configurations (config) from Lekuka.



2.2. Device communication modes

Device communicates with E-Invoicing Gateway API in one of two possible communication modes:

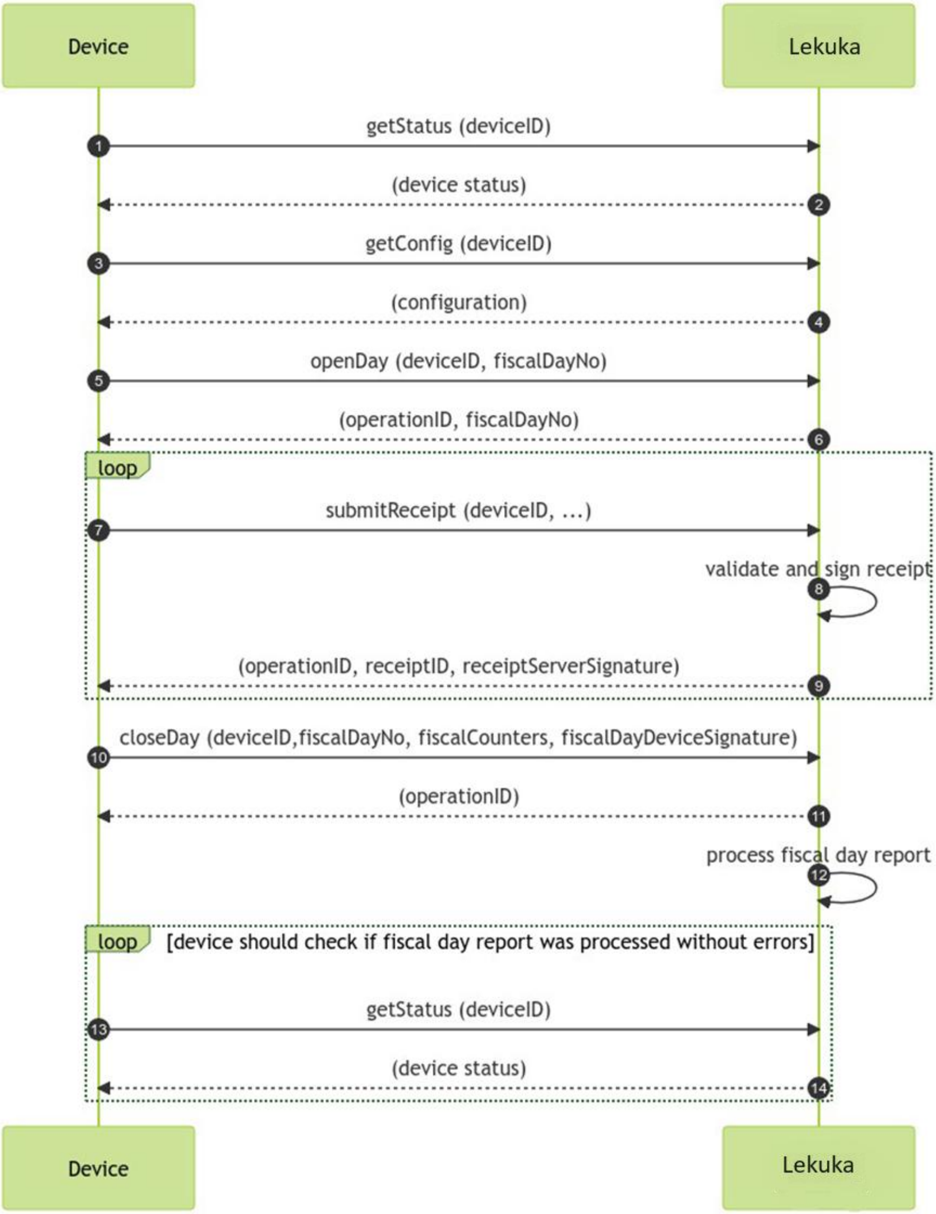
- Online
- Offline

Online communication mode represents device communication in a way when device must have online access to Lekuka (must have internet connection available) when it wants to close fiscal period. Fiscal period opening and submission of an invoice to Lekuka should be done immediately after opening a fiscal period or printing receipt or invoice for buyer respectively, however in case of missing internet connection, fiscal period opening message and submission of a receipt may be delayed but must be done before closing a fiscal period. In case fiscal period was opened, and invoice was issued without internet connection, it is mandatory, that fiscal period opening message would be sent before sending invoices. Otherwise, invoices will not be accepted.

Offline communication mode represents device communication in a way, when device may not have internet, and its receipts and report data will be provided to Lekuka by using files (by uploading file using Self-service, or by sending file to E-Invoicing Gateway API , whenever connection will be available).

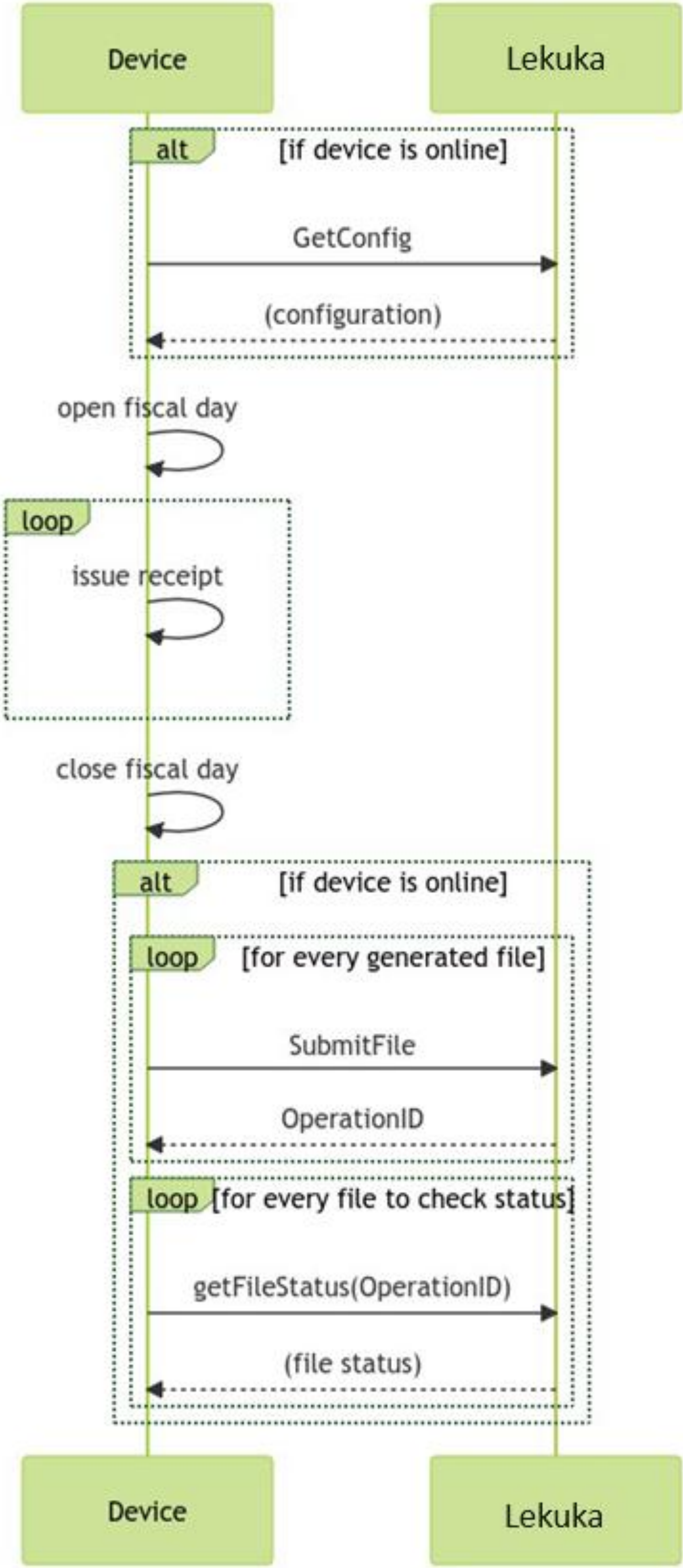
2.3. Fiscal period (in online mode)

After successful device registration, it can be used for submitting sales to Lekuka. Sales submission is possible only when fiscal period is opened. When work is finished with device, it must close fiscal period.



In case of error in fiscal period report processing report must be corrected on device and resubmitted to Lekuka. Report resubmission can be done unlimited number of times. In case it does not give successful result, and supplier cannot fix it to submit successful report, fiscal period may be closed manually by RSL officer.

2.4. Fiscal period (in offline mode)



If device has internet connection, it should call getConfig method, to get device configuration information.

Device opens fiscal period, if device has internet connection it can send information using submitFile method (file with only header) about opened day.

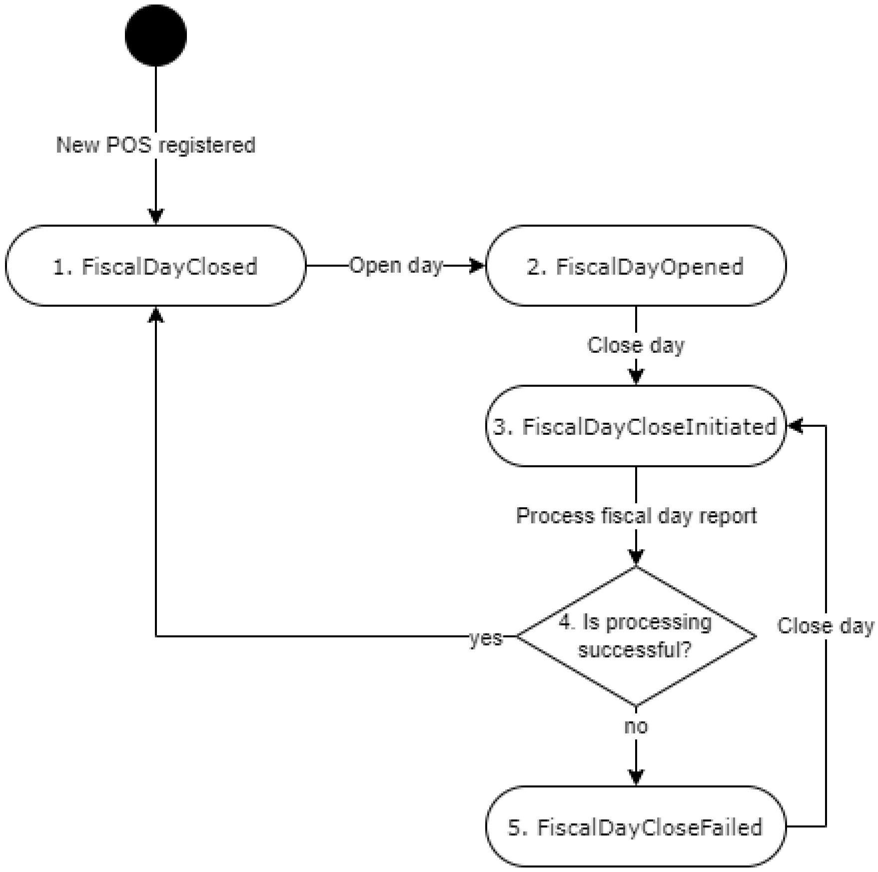
Invoices are issued which are saved in device if device has internet connection it can send information using submitFile method (file with header and content) about already issued invoices.

After fiscal period is closed information is saved in device, if device has internet connection it should send information using submitFile method (file with header and footer) about closed day. However, if there are still left unsent invoices it should send information using submitFile method (file with header, content and footer) about receipts and closed day.

After sending file device should call getFileStatus method to get sent file status.

3. OBJECT STATUSES

3.1. Fiscal period statuses



Status	Description
1. FiscalDayClosed	Status used when device has successfully closed fiscal period. New fiscal period opening is possible only from this status.
2. FiscalDayOpened	Fiscal period is opened. Invoices can be created only when fiscal period is in this status.
3. FiscalDayCloseInitiated	Closure of fiscal period is initiated, however Lekuka not yet validated counters. While fiscal period is in status “FiscalDayCloseInitiated”, no new request to initiate fiscal period closure will be accepted from device or user. New invoices will not be accepted as well.
4. Is processing successful?	If processing of report is successful, fiscal period status is changed to FiscalDayClosed, if processing of report is not successful status is changed to FiscalDayCloseFailed.
5. FiscalDayCloseFailed	Lekuka validated fiscal period report received from device, however there are validation errors. Device must correct issues and repeatedly submit fiscal period closure message.

4. E-INVOICING GATEWAY API INTERFACES

E-Invoicing Gateway API exposes its methods using REST JSON interface. All methods except closeDay and submitFile are synchronous. closeDay and submitFile methods return response about accepted request synchronously, however processing of information is done asynchronously.

Each request must contain these HTTP headers:

Header name	Mandatory	Description
DeviceModelName	Yes	Device model name as registered in RSL
DeviceModelVersionNo	Yes	Device model version number as registered in RSL

During processing of request validations are performed and these errors may be returned:

- If device model is blacklisted error DEV04 is returned.
- If taxpayer is not active error DEV05 is returned.
- If device is not active error DEV01 is returned.

4.1. verifyTaxpayerInformation

verifyTaxpayerInformation (/Public/v1/{deviceID}/VerifyTaxpayerInformation) endpoint is used to retrieve taxpayer information from Lekuka before device registration (in order user could double check if device is going to be registered to correct taxpayer). Activation key is not yet used.

This API endpoint does not require certificate for authentication.

Input parameters:

Name	Type	Mandatory	Description
deviceID	Int	Yes	Device ID
activationKey	String (8)	Yes	Activation key.
deviceSerialNo	String (20)	Yes	Device serial number assigned by manufacturer.

Output parameters:

Name	Type	Mandatory	Description
operationID	String (60)	Yes	Operation ID assigned by Lekuka.
taxPayerName	String (250)	Yes	Taxpayer name
taxPayerTIN	String (11)	Yes	Taxpayer TIN code
vatNumber	String (9)	No	Taxpayer's VAT number. Field is not returned if taxpayer is not a VAT payer.
deviceBranchName	String (250)	Yes	Device branch name (or trade name) assigned by taxpayer.
deviceBranchAddress	Address	Yes	Device branch address.
deviceBranchContacts	Contacts	No	Device branch contacts information.

4.2. registerDevice

registerDevice (/Public/v1/{deviceID}/RegisterDevice) endpoint is used to get device certificate and register device in Lekuka (link device with Lekuka).

This API endpoint does not require certificate for authentication.

Input parameters:

Name	Type	Mandatory	Description
deviceID	Int	Yes	Device ID
activationKey	String (8)	Yes	Activation key. Case insensitive 8 symbols key.
certificateRequest	String	Yes	Certificate signing request (CSR) for which certificate will be generated (in PEM format). Assigned by RSL device name (format: RSL-<Fiscal_device_serial_no>-<zero_padded_10_digit_deviceId>) should be provided in CSR`s Subject Example of CN value, when device serial no is “SN: 001” and device id is “187”: “RSL-SN: 001-0000000187”. Other CSR’s Subject fields are optional, however if provided must match these values (otherwise device registration will be rejected): • C = LS • O = Revenue Services Lesotho • S = Lesotho Supported algorithms and key types (in order of suggested preference): 1) ECC ECDSA on SECG secp256r1 curve (also named as ANSI prime256v1, NIST P-256); Signature Algorithm: ecdsa-with-SHA256. 2) RSA 2048; Signature Algorithm - SHA256WithRSA. *Note:* RSA 2k and ECC 256 implement the same security level, which is considered safe until 2030 year (considering trends due to upgrade in computer software and hardware combination). Most cryptographic tools and libraries support this format giving easy-to-use API and hiding all technical representation and encoding details. CSR is “CertificationRequest” structure, as defined by PKCS #10 (CSR syntax specified by RFC2986). Serialized in PEM format (i.e., base64 encoded with “------BEGIN CERTIFICATE REQUEST-----” header and “----END CERTIFICATE REQUEST-----” footer). For more info, please refer to “12 Certificate signing request (CSR) and Certificate examples”.

Output parameters:

Name	Type	Mandatory	Description
operationID	String (60)	Yes	Operation ID assigned by Lekuka.
certificate	String	Yes	X.509 v3 type device certificate (in PEM format). It must be used by device in further communication with E-Invoicing Gateway API. Certificate is multi-purpose: • Client Certificate for SSL with Client Authentication. • For data signing when device signature is required. Certificate is “Certificate” structure specified by RFC5280. Serialized in PEM format (i.e. base64 encoded with “-----BEGIN CERTIFICATE-----” header and “----END CERTIFICATE-----” footer). For more info, please refer to “12 Certificate signing request (CSR) and Certificate examples”.

4.3. issueCertificate

issueCertificate (/Device/v2/{deviceID}/IssueCertificate) endpoint is used to renew certificate before the expiration of the current certificate.

It is recommended to renew certificate a month before its expiration.

Certificate reissuance can be done at any time. It does not depend on fiscal period status, however it is recommended to be done before opening a new fiscal period.

Input parameters:

Name	Type	Mandatory	Description
deviceID	Int	Yes	Device ID
certificateRequest	String	Yes	Certificate signing request (CSR) for which certificate will be generated (in PEM format). certificateRequest requirements are specified in registerDevice endpoint description.
twoStepsProcess	Boolean	Yes	Specifies if new certificate issuance will be done in a single step, or as a two steps process where device will additionally confirm that new certificate is successfully received. Possible values: - True – two steps will be used where device will confirm certificate issuance by calling confirmCertificate method - False - single step for certificate change will be used, where device will not additionally confirm certificate issuance If parameter value is set to false, new certificate is valid from its generation moment and previous certificate is revoked immediately. If parameter value is set to true, new certificate will be used only after receiving confirmCertificate request from device. Unti then previous certificate will be used (it will be revoked after confirmCertificate is received). If parameter value is set true, but instead of receiving confirmCertificate request, new issueCertificate request is received certificate generated with previous issueCertificate request will be revoked and a new one is generated.

Output parameters:

Name	Type	Mandatory	Description
operationID	String (60)	Yes	Operation ID assigned by Lekuka.
certificate	String	Yes	X.509 v3 type device certificate (in PEM format). Certificate requirements are specified in registerDevice endpoint description.

4.4. confirmCertificate

confirmCertificate (/Device/v2/{deviceID}/ConfirmCertificate) endpoint is used to confirm that new certificate is received.

It is mandatory to confirm after each certificate renewal.

Input parameters:

Name	Type	Mandatory	Description
deviceID	Int	Yes	Device ID
thumbprint	Binary (20)	Yes	Thumbprint of Lekuka signing certificate which is confirmed.

Output parameters:

Name	Type	Mandatory	Description
operationID	String (60)	Yes	Operation ID assigned by Lekuka.

4.5. getConfig

getConfig (/Device/v2/{deviceID}/GetConfig) endpoint is used to retrieve taxpayers and device information and configuration.

Input parameters:

Name	Type	Mandatory	Description
deviceID	Int	Yes	Device ID

Output parameters:

Name	Type	Mandatory	Description
operationID	String (60)	Yes	Operation ID assigned by Lekuka.
taxPayerName	String (250)	Yes	Taxpayer name
taxPayerTIN	String (11)	Yes	Taxpayer TIN code
vatNumber	String (9)	No	Taxpayer's VAT number. Field is not returned if taxpayer is not a VAT payer. If taxpayer which is not a VAT taxpayer, gets VAT number and its device has opened fiscal period, fiscal period must be closed and newly opened in order taxpayer could submit receipts with VAT.
deviceSerialNo	String (20)	Yes	Device serial number assigned by manufacturer.
deviceBranchName	String (250)	Yes	Device branch name (or trade name) assigned by taxpayer.
deviceBranchAddress	Address	Yes	Device branch address.
deviceBranchContacts	Contacts	No	Device branch contacts information.
deviceOperatingMode	DeviceOperatingMode	Yes	Specifies what are allowed receipt processing modes for this device. Possible values: - Online - Offline Device operational mode can be changed only by RSL officer.
taxPayerDayMaxHrs	Int	Yes	Maximum fiscal period duration in hours.
taxpayerDayEndNotificationHrs	Int	Yes	How much time in hours before end of fiscal period device should show notification to salesperson.
applicableTaxes	Tax array	Yes	List of applicable tax rates which can be used by this taxpayer and are valid during getConfig request time or will be valid in the future.
certificateValidTill	Date	Yes	Date till when device certificate is valid. Device must reissue new certificate before this date. After this date device will not be able to submit any request to E-Invoicing Gateway.
qrUrl	String (50)	Yes	URL for QR preparation. This URL needs to be used by device when generating QR code printed on invoice.

Tax:

Name	Type	Mandatory	Description
taxID	Int	Yes	Tax ID uniquely identifying a tax. This tax ID must be used in submitting invoices.
taxRate	Decimal (10,2)	No	Tax percentage or fixed value. In case of exempt, field will not be returned.
taxName	String (50)	Yes	Tax name.
taxValidFrom	Date	Yes	Date from which tax is valid.
taxValidTill	Date	No	Date till which tax is valid.
taxType	TaxType	Yes	Type of tax.

4.6. getStatus

getStatus (/Device/v1/{deviceID}/GetStatus) endpoint is used to get fiscal period status.

Request can't be sent if DeviceOperatingMode is Offline. If DeviceOperatingMode is Offline error DEV01 is received.

Input parameters:

Name	Type	Mandatory	Description
deviceID	Int	Yes	Device ID

Output parameters:

Name	Type	Mandatory	Description
operationID	String (60)	Yes	Operation ID assigned by Lekuka.
fiscalDayStatus	FiscalDayStatus	Yes	Device fiscal period status.
fiscalDayReconciliationMode	FiscalDayReconciliationMode	No	In case fiscal period status is "FiscalDayClosed" defines how it was closed: automatically or manually.
fiscalDayServerSignature	SignatureDataEx	No	Fiscal period report signature prepared by Lekuka. This field is returned only when fiscalDaySatus is "FiscalDayClosed". This signature is not used in further communication or any data preparation for Lekuka. It is confirmation from Lekuka that fiscal period is closed and should be stored on device. Signature verification rules are described in section 13.3.
fiscalDayClosed	DateTime	No	Date and time when fiscal period report was processed, and fiscal period status was changed to "FiscalDayClosed". Time is provided in local time without time zone information. This field is returned only when fiscalDaySatus is "FiscalDayClosed". If device has never started a new fiscal period, this field is not returned.
fiscalDayClosingErrorCode	FiscalDayProcessingError	No	Code of error which appears during fiscal period closure. Possible codes are defined in section 5.4.10 FiscalDayProcessingError. This field is returned only when fiscalDayStatus is "FiscalDayCloseFailed".
fiscalDayCounters	FiscalDayCounter array	No	List of fiscal period counters. This field is returned only when fiscalDayStatus is "FiscalDayClosed" and fiscalDayReconciliationMode is "Manual". List contains only non-zero value counters. FiscalDayCounter type description provided in *closeDay* endpoint.
fiscalDayDocumentQuantities	FiscalDayDocumentQuantity array	No	List of fiscal period document quantities. This field is returned only when fiscalDayStatus is "FiscalDayClosed" and fiscalDayReconciliationMode is "Manual". FiscalDayDocumentQuantity type description provided in FiscalDayDocumentQuantity table.
lastReceiptGlobalNo	Int	No	Last submitted receiptGlobalNo field value of invoice, credit note, or debit note. In case no document is yet submitted from this device, this field is not returned.
lastFiscalDayNo	Int	No	In case fiscal period is opened, current fiscal period fiscalDayNo is returned. In case fiscal

			period is closed, last closed fiscal period fiscalDayNo is returned. In case device is new and not yet opened its first fiscal period, this field is not returned.

FiscalDayDocumentQuantity:

Name	Type	Mandatory	Description
receiptType	ReceiptType	Yes	Type of receipt.
receiptCurrency	String (3)	Yes	Receipt currency (ISO 4217 currency code).
receiptQuantity	Int	Yes	Total quantity of receipts of particular receipt type and currency for fiscal period.
receiptTotalAmount	Decimal (19,2)	Yes	Total receipt amount (including tax) of receipts of particular receipt type and currency for fiscal period.

4.7. openDay

openDay (/Device/v1/{deviceID}/OpenDay) endpoint is used to open a new fiscal period. Opening of new fiscal period is possible only when previous fiscal period is successfully closed (fiscal period status is "FiscalDayClosed"). Opening of a new fiscal period in a device may be done without internet connection. It is important that such delayed request about day opening is sent before sending receipts.

Request can't be sent if DeviceOperatingMode is Offline. If DeviceOperatingMode is Offline error DEV01 is received.

Input parameters:

Name	Type	Mandatory	Description
deviceID	Int	Yes	Device ID
fiscalDayOpened	DateTime	Yes	Date and time when fiscal period was opened on a device. Time is provided in local time without time zone information.
fiscalDayNo	Int	No	Fiscal period number assigned by device. If this field is not sent, Lekuka will generate fiscal period number and return it to device. **Validation rules:** - fiscalDayNo must be equal to 1 for the first fiscal period of device - fiscalDayNo must be greater by one from the last closed fiscal period fiscalDayNo.

Output parameters:

Name	Type	Mandatory	Description
operationID	String (60)	Yes	Operation ID assigned by Lekuka.
fiscalDayNo	Int	Yes	Fiscal period number of opened day. In case device has sent fiscalDayNo in request, it is returned in this field. In case device has not sent it, new fiscal period number will be generated by Lekuka.

4.8. submitReceipt

submitReceipt (/Device/v2/{deviceID}/SubmitReceipt) endpoint is used to submit a receipt to Lekuka in online mode and get a Lekuka signature for it (signature is not a QR code, it is an acknowledgement of Lekuka about received receipt). Receipt can be submitted only when fiscal period status is "FiscalDayOpened" or "FiscalDayCloseFailed".

Request can't be sent if DeviceOperatingMode is Offline. If DeviceOperatingMode is Offline error DEV01 is received.

In case device tried to close a fiscal, period and attempt was unsuccessful, device still have a possibility to submit a new receipt. In case it fails to submit, there should be retries of the unsent invoices to be sent.

In case the same receipt (with the same deviceID, receiptGlobalNo and receiptHash) is submitted more than once, E-Invoicing Gateway API will return successful result to device with the same original receipt receiptID, receiptServerSignature, however different operationID.

Each submitted receipt is validated. Receipt will not be accepted, error will be returned to device (as specified in 8.2 Error codes), in these cases:

- device status is other than "*Active*";
- fiscal period status is other than "*FiscalDayOpened*" or "*FiscalDayCloseFailed*";
- receipt message structure is not valid.

In case the above-mentioned validations have passed, but receipt has other validation issues specified below (described in "Validation rules"), receipt will be accepted and signed, but will be marked as invalid with validation color code assigned (as specified in 8.3. Validation errors).

Each submitted receipt, must increase fiscal period counters as specified in *5.4.14. TaxRoundingType*

Type of tax. Possible values:

Enum value	Enum order
PerReceipt	0
PerReceiptLine	1

Counters.

Input parameters:

Name	Type	Mandatory	Description
deviceID	Int	Yes	Device ID
receipt	Receipt	Yes	Receipt data

Receipt:

Name	Type	Mandatory	Description
receiptType	ReceiptType	Yes	Type of receipt.
receiptCurrency	String (3)	Yes	Receipt currency (ISO 4217 currency code). Must be LSL. **Validation rules** RCPT010: currency code must be present in Lekuka and must be valid at the time of receiptDate.
receiptCounter	Int	Yes	Daily ascending serial number of receipt assigned by taxpayer's device. **Validation rules:** RCPT011: receiptCounter must be equal to 1 for the first document in fiscal period and receiptCounter must be greater by one from the previous document receiptCounter value for the second and other document in fiscal period. RCPT052: receiptCounter must be unique in current fiscal period.
receiptGlobalNo	Int	Yes	Cumulative ascending serial number of total documents issued since device activation date.

			Taxpayer is allowed to reset this receiptGlobalNo counter to start from 1, however this is allowed to be done only for the first document in a fiscal period. **Validation rules:** RCPT012: receiptGlobalNo must be greater by one from the previous document receiptGlobalNo or may be equal to 1 for the first document in fiscal period. RCPT053: receiptGlobalNo must be unique in current fiscal period.
invoiceNo	String (50)	Yes	Invoice number generated by taxpayers business system (accounting, ERP, cashregister or any other system). **Validation rules:** RCPT013: invoiceNo must be unique in taxpayer context.
buyerData	Buyer	No*	Buyer information. Field is mandatory when receiptType is FiscalInvoice or any other type where receiptTotal is bigger than ThresholdForMandatoryBuyerData of LSL. **Validation rules:** RCPT046RCPT043RCPT013: -buyerData is mandatory for provided document
receiptNotes	String (500)	No*	Additional notes for provided document. Usually used in CreditNote and DebitNote document to indicate a reason for document issuance. Mandatory for CreditNote and DebitNote. **Validation rules:** RCPT034: - receiptNotes is mandatory for receiptType CreditNote and DebitNote
receiptDate	DateTime	Yes	Date and time of device when documents is printed for customer. Time is provided in local time without time zone information. **Validation rules:** RCPT014: - receiptDate must be greater than fiscal period opening date and time RCPT030: - receiptDate must be greater than previously submitted document receiptDate RCPT031: - receiptDate must not be greater than current time (time difference set in AllowedTimeDifferenceForReceiptSubmission setting is allowed). RCPT041: - receiptDate must be less or equal than fiscal period opened + taxpayerDayMaxHrs.
creditDebitNote	CreditDebitNote	No*	Information of original document (Receipt, FiscalInvoice or Payout) which is Credited or debited by this document. This field is mandatory in case receipt type is CreditNote or DebitNote. **Validation rules:** RCPT015: - creditDebitNote object is mandatory for receiptType CreditNote and DebitNote RCPT032: - credited or debited document must exist in Lekuka - original document which is credited or debited must be submitted by the same taxpayer as this document RCPT033: - original document which is credited or debited by this document must be issued not earlier than 12 months before credit or debit note receiptDate RCPT035: - total credit note amount must not exceed original document amount with all previously submitted credit

			and debit notes amounts, where amount is calculated in this way (using absolute values): original document amount - all submitted credit notes amounts + all submitted debit notes amounts. RCPT036: - credit or debit note may have all or part of taxes (percentages plus tax ids) used in the original document. It cannot contain new taxes, that where not submitted in original document (example, if original document has exempt and 15% VAT tax lines, credit or debit note may have only exempt, 15% VAT or both tax lines, but cannot have 0% tax line). Non-VAT taxpayer can still send VAT tax line if such line was present on original document. RCPT029: - CreditDebitNote object must not be provided for FiscalInvoice / Receipt / Payout document. RCPT042: - currency code must be same as in the original document. Credit/debit note cannot have different currency, than in original invoice.
receiptLinesTaxInclusive	Boolean	Yes	This field specifies if provided document's receiptLines are tax inclusive or not. Possible values: - True, all receipt lines are tax inclusive - False, all receipt lines are tax exclusive
taxRoundingType	TaxRoundingType	No	Indicates the method how tax is calculated in receipt. Tax can be calculated in one of these methods: • PerReceipt – it means that all receipt lines of the same tax are summed, tax is calculated and result is rounded to 2 decimal places; • PerReceiptLine – it means that tax is calculated for each of receipt line, each line tax is rounded to 2 decimal places and then summed. Default value if field is not sent – PerReceipt.
receiptLines	ReceiptLine array	Yes	Document lines. **Validation rules:** RCPT016: at least one line must be provided.
receiptTaxes	ReceiptTax array	Yes	Applied taxes aggregated per taxId and taxCode (provided separate line for each different taxId and taxCode combination used in receiptLines). **Validation rules:** RCPT017: at least one line must be provided.
receiptPayments	Payment array	Yes	Means of payments how document was paid. **Validation rules:** RCPT018: at least one line must be provided.
receiptTotal	Decimal (19,2)	Yes	Total document amount which is paid/received by buyer. **Validation rules:** RCPT019: - for document which is tax inclusive (receiptLinesTaxInclusive is true) receiptTotal must be equal to sum of receiptLineTotal of all receiptLines. RCPT037: - for document which is tax exclusive (receiptLinesTaxInclusive is false) receiptTotal must be equal to sum of receiptLineTotal of all receiptLines plus sum of taxAmount of all receiptTaxes. RCPT038RCPT038RCPT038RCPT038RCPT038: - receiptTotal must be equal to sum of salesAmountWithTax of all taxLines where taxType is VAT, NonVAT and Exempt plus sum of taxAmount of all taxLines where taxType is WithholdingTax, FixedValueLevy, PercentageLevy. RCPT039: - receiptTotal must be equal to sum of paymentAmount of all receiptPayments. RCPT040:

			- receiptTotal must be greater than or equal to 0 for Receipt, FiscalInvoice, - receiptTotal must be greater than or equal to 0 for DebitNote if it is created for Receipt or FiscalInvoice document - receiptTotal must be greater than or equal to 0 for CreditNote if it is created for Payout document; - receiptTotal must be less than or equal to 0 for Payout, - receiptTotal must be less than or equal to 0 for DebitNote if it is created for Payout document - receiptTotal must be less than or equal to 0 for CreditNote if it is created for Receipt or FiscalInvoice document;
receiptPrintForm	ReceiptPrintForm	No	The format in which document is printed and delivered to buyer (as a receipt or on A4 paper). Default value if field is not sent: Receipt48.
receiptDeviceSignature	SignatureData	Yes	SignatureData structure with SHA256 hash of document fields (hash used for signature) and receipt device signature prepared by using device private key as described in section 13.2. Signature must be prepared with certificate which was valid during receiptDate moment. **Validation rules:** RCPT020: receiptDeviceSignature must be correct

Buyer:

Name	Type	Mandatory	Description
buyerRegisterName	String (250)	Yes	Buyer company name or physical person name and surname. RCPT043: buyerRegisterName and buyerTIN are mandatory fields if buyer data object is sent.
buyerTradeName	String (250)	No	Buyer trade name (store name, or branch name).
buyerTIN	String (11)	Yes	Buyer TIN. **Validation rules:** RCPT043: buyerRegisterName and buyerTIN are mandatory fields if buyer data object is sent. RCPT054: buyerTIN field value must be different from Taxpayer TIN who is issuing an invoice.
VATNumber	String (8)	No	Buyer VAT number.
buyerContacts	Contacts	No	Buyer contacts.
buyerAddress	Address	No	Buyer address.

CreditDebitNote:

Name	Type	Mandatory	Description
receiptID	Bigint	No	Receipt ID of original document which is credited or debited by current document. receiptID must be sent or deviceID with receiptGlobalNo and fiscalDayNo must be sent.
deviceID	Int	No	Device ID of original document which is credited or debited by current document. In case receiptID is sent, this field is ignored.
receiptGlobalNo	Int	No	Receipt global No of original document which is credited or debited by current receipt. In case receiptID is sent, this field is ignored.
fiscalDayNo	Int	No	fiscalDayNo of original document which is credited or debited by current document.

ReceiptLine:

Name	Type	Mandatory	Description
receiptLineType	ReceiptLineType	Yes	Type of receipt line (for example sales or discount). **Validation rules:** RCPT044: Receipt line type payout only allowed in payout document or debit/credit note document if credit or debit note is created for payout document RCPT045: Receipt line types discount and sale are not allowed in payout document and debit/credit note document if credit or debit note is created for payout document.
receiptLineSubType	ReceiptLineSubType	No	Subtype of receipt line (for example product or services).
receiptLineNo	Int	Yes	Line sequence number in receipt.
receiptLineHSCode	String (8)	No	Product or service code.
receiptLineName	String (200)	Yes	Product or service name.
receiptLinePrice	Decimal (19,3)	No	Price of product or service in receipt currency (for single item). It may not be provided when taxpayer set a special price for bundle of items (i.e., when selling 3 items for 1 LSL). If receipt lines are tax inclusive receiptLinePrice besides of item price without tax must also include all applicable taxes. **Validation rules:** RCPT022: receiptLinePrice value must be: - greater than 0 for receiptType FiscalInvoice/Receipt/DebitNote if ReceiptLineType is Sale; - less than 0 for receiptType CreditNote if ReceiptLineType is Sale - less than 0 for receiptType FiscalInvoice/Receipt if ReceiptLineType is Discount; - greater than 0 for receiptType CreditNote if ReceiptLineType is Payout; - less than 0 for receiptType DebitNote if ReceiptLineType is Payout; - less than 0 for receiptType Payout if ReceiptLineType is Payout;
receiptLineQuantity	Decimal (19,3)	Yes	Product quantity **Validation rules:** RCPT023: value must be greater than 0
receiptLineTotal	Decimal (19,2)	Yes	Total price of document line (receiptLinePrice * receiptLineQuantity). If receiptLinesTaxInclusive is true, total price includes all applicable taxes. **Validation rules:** RCPT022: in case receiptLinePrice is not provided receiptLineTotal value bus be: - greater than 0 for receiptType FiscalInvoice/Receipt/DebitNote if ReceiptLineType is Sale; - less than 0 for receiptType CreditNote if ReceiptLineType is Sale - less than 0 for receiptType FiscalInvoice/Receipt if ReceiptLineType is Discount; - greater than 0 for receiptType CreditNote if ReceiptLineType is Payout; - less than 0 for receiptType DebitNote if ReceiptLineType is Payout; - less than 0 for receiptType Payout if ReceiptLineType is Payout; RCPT024: in case receiptLinePrice is provided, receiptLineTotal must be equal to receiptLinePrice * receiptLineQuantity (arithmetical rounding is applied)
taxCode	String (3)	No	Tax code representation in receipt. This field is not mandatory; however, it must be provided for all receipt lines or none of them. It is not allowed to provide taxCode for just a part of lines.

taxRate	Decimal (10,2)	No	Tax rate applied to receipt line (VAT or NonVAT tax percent). In case of exempt tax, this field should not be provided. **Validation rules:** RCPT025: - Tax rate and tax ID combination must be the same as in Lekuka.
taxID	Int	Yes	Applied tax ID uniquely identifying used tax. **Validation rules:** RCPT025: - tax ID value must be one of the allowed tax ID values - receiptDate must be in a period of tax valid from and valid till period RCPT021: - non-VAT taxpayer can use only NonVAT or exempt tax types. RCPT050: - Only these types of taxes are allowed to be sent as main taxes: VAT or NonVAT or Exempt
additionalTaxes	AdditionalTax array	No	Additionally applied taxes. This array is mandatory if product or service has applied levies, additional taxes.

ReceiptTax:

Name	Type	Mandatory	Description
taxCode	String (3)	No	Tax code representation in receipt.
taxRate	Decimal (10,2)	No	Applied tax percentage or fixed value: - In case of non-VAT sale, 0 value should be used; - In case of exempt this field should not be provided. **Validation rules:** RCPT025: -Tax rate, tax type and tax ID combination must be the same as in Lekuka.
taxID	Int	Yes	Applied tax ID uniquely identifying used tax. **Validation rules:** RCPT025: - tax ID value must be one of the allowed tax ID values - receiptDate must be in a period of tax valid from and valid till period RCPT021: - non-VAT taxpayer can use only NonVAT, Exempt, FixedValueLevy or PercentageLevy tax types. RCPT049: - Receipt taxes must include all the taxes sent in receiptLines (indicated in receiptLine/taxID as wel as in receiptLine/additionalTaxes/taxId)
taxType	TaxType	Yes	Type of tax.
taxAmount	Decimal (19,2)	Yes	Total amount of payable tax for this tax line. Tax can be calculated in one of these ways: 1) **Per receipt line** (taxRoundingType = PerReceiptLine) a) Tax is calculated for each receipt line b) Calculated tax result of each line is rounded up to 2 decimal places c) Each line tax is summed up by taxID and taxCode (this is the result sent to Lekuka) 2) **Grouped** (taxRoundingType = PerReceipt) a) All receipt lines are summed up by taxId and taxCode b) Tax is calculated from aggregated amount c) Result is rounded up to 2 decimal places (this is the result sent to Lekuka) Tax amount calculation formula: 1. When tax type is Exempt, taxAmount = 0. 2. When tax type is FixedValueLevy, taxAmount = appliedForQuantity (where it is indicated) or receiptLineQuantity (where appliedForQuantity is not sent) multiplied by taxRate. 3. When tax type is VAT, NonVAT or PercentageLevy:

			3.1. When receiptLinesTaxInclusive = true (it means receiptLineTotal amount is inclusive all applied taxes), taxAmount is calculated in this way: 3.1.1. When taxRoundingType = PerReceipt: taxAmount = ROUND((receiptLineTotal - FixedValueLevy) / (100 + all percentage tax rates) * taxRate, 2) 3.1.2. When taxRoundingType = PerReceiptLine: taxAmount = ROUND(ROUND((receiptLineTotal - FixedValueLevy) * (receiptLineTotal - FixedValueLevy) / ROUND((receiptLineTotal - FixedValueLevy) * (100 + all percentage tax rates), 2), 2) * taxRate, 2) 3.2. When receiptLinesTaxInclusive = false (it means receiptLineTotal amount is before taxes), taxAmount = receiptLineTotal multiplied by taxRate. **Validation rules:** RCPT026: - This error is returned in case received taxAmount is not equal to tax calculation result based on indicated taxRoundingType.
salesAmountWithTax	Decimal (19,2)	Yes	Total sales amount (including tax) for this tax line. As tax can be calculated in two different ways (per line or grouped) as described in detail above (in taxAmount field description), respectively salesAmountWithTax may slightly differ. Sales amount with tax calculation formula: 1. When receiptLinesTaxInclusive = true (it means receiptLineTotal amount is inclusive all applied taxes): 1.1. If receiptLine has a single tax applied (without any additional tax), salesAmountWithTax = receiptLineTotal 1.2. If receiptLine has more than a single tax applied, then all taxes need to be removed to get sales amount without tax and taxAmount needs to be added to sales amount without tax. 1.2.1. When taxRoundingType = PerReceipt: salesAmountWithTax = ROUND((receiptLineTotal – FixedValueLevy) / (100 + all percentage tax rates) * (100+ taxRate), 2) 1.2.2. When taxRoundingType = PerReceiptLine: salesAmountWithTax = receiptLineTotal – SUM (all receiptTax/taxAmount) + taxAmount of specific tax line. 2. When receiptLinesTaxInclusive = false (it means receiptLineTotal amount is before taxes): 2.1. For exempt, VAT, NonVAT and PercentageLevy salesAmountWithTax = ROUND(receiptLineTotal multiplied by (100 + taxRate) / 100, 2) 2.2. For FixedValueLevy salesAmounWithTax = taxRate multiplied by appliedForQuantity (where it is indicated) or receiptLineQuantity (where appliedForQuantity is not sent) + receiptLineTotal. **Validation rules:** RCPT027: - This error is returned in case received salesAmountWithTax is not equal to the above calculation result (for receipts where receiptLinesTaxInclusive = false, salesAmountWithTax calculation is checked based on indicated taxRoundingType).

Payment:

Name	Type	Mandatory	Description
moneyTypeCode	MoneyType	Yes	Code of payment means by which payment was done.
paymentAmount	Decimal (19,2)	Yes	Amount paid by this payment type in receipt currency. In case customer gave bigger amount (bill) in cash than a total amount to be pay, it needs to be sent amount without change to buyer. **Validation rules:** RCPT028:

			- paymentAmount must be greater than or equal to 0 for Receipt, FiscalInvoice - paymentAmount must be greater than or equal to 0 for DebitNote if it is created from Receipt or FiscalInvoice document - paymentAmount must be greater than or equal to 0 for CreditNote if it is created from Payout document; - paymentAmount must be less than or equal to 0 for Payout - paymentAmount must be less than or equal to 0 for DebitNote if it is created from Payout - paymentAmount must be less than or equal to 0 for CreditNote if it is created from Receipt or FiscalInvoice document;

AdditionalTax:

Name	Type	Mandatory	Description
taxID	Int	Yes	Applied tax ID uniquely identifying used tax. **Validation rules:** RCPT025: - tax ID value must be one of the allowed tax ID values - receiptDate must be in a period of tax valid from and valid till period RCPT050: -Only these types of taxes are allowed to be sent as additional taxes: FixedValueLevy or PercentageLevy or WithholdingTax
receiptLineId	Int	Yes	Receipt line id.
taxCode	String (3)	No	Tax code representation in receipt.
taxType	TaxType	Yes	Type of tax.
taxRate	Decimal (10,2)	Yes	Applied tax percentage or fixed value levy rate. **Validation rules:** RCPT025: -Tax rate, tax type and tax ID combination must be the same as in Lekuka.
appliedForQuantity	Int	No	Quantity for which FixedValueLevy is applied. It is used for FixedValueLevy if item is sold in packages, but levy is applied per unit. This amount is describing for how many units levy is applied. Mandatory field if TaxType is FixedValueLevy. RCPT051RCPT051RCPT051RCPT043RCPT013: - appliedForQuantity is mandatory for TaxType FixedValueLevy

Output parameters:

Name	Type	Mandatory	Description
operationID	String (60)	Yes	Operation ID assigned by Lekuka.
receiptID	Bigint	Yes	Receipt ID assigned by Lekuka.
serverDate	DateTime	Yes	Date and time when Lekuka signed a receipt.
receiptServerSignature	SignatureDataEx	Yes	Receipt Lekuka signature generated by Lekuka. This signature is not used in further communication or any data preparation for Lekuka. It is confirmation from Lekuka that receipt is accepted and should be stored on device. Signature verification rules are described in section 13.2.
validationErrors	ValidationError array	No	Invoice validation errors found by Lekuka. Returned if invoice has at least one validation error described in: 8.3 Validation errors

ValidationError:

Name	Type	Mandatory	Description
validationErrorCode	String (10)	Yes	Validation error code.
validationErrorColor	String (10)	Yes	Validation error color.

4.9. submitFile

submitFile (/Device/v1/{deviceID}/SubmitFile) endpoint is used to submit a batch of invoices to Lekuka in a single file.

Request can be sent if DeviceOperatingMode is Offline.

File can have three parts: Header (fiscal period day information), Content (invoices information) and Footer (Z report information). Header is always mandatory, Content and Footer are optional (Footer can be sent without Content, Content can be sent without Footer). File must have information only from a single fiscal period (single file cannot contain receipts from different fiscal periods). Information must be sent only for closed fiscal period. Elements sequence in file must be assured to be Header, Content, Footer, otherwise file format error will be returned. Footer must be sent only in last file for particular fiscal period.

File must be JSON format sent as base64 encoded string.

File will not be accepted, error will be returned to device (as specified in 8.2 Error codes), in these cases:

- device operating mode is other than “Offline”;
- file is bigger than 3 MB;
- request structure is not valid;
- device id in input parameters and file header are different;
- file Header failed to be parsed;
- file is sent for already closed day.

In case the above-mentioned validations have passed, file is processed asynchronously.

File status can be received by calling 4.10 getFileStatus API method.

If other validation issues during asynchronous processing will be detected error information may be retrieved in 4.10 getFileStatus response.

During asynchronous processing these steps will be executed:

- file structure will be validated;
- fiscal period day status validated by information provided in header;
- invoices will be imported from file to Lekuka (same validation rules will be applied as described in **Error! Reference source not found. Error! Reference source not found.**);
- Z report will be validated same validation rules will be applied as described in **Error! Reference source not found. Error! Reference source not found.**).

Additional validation rules:

- If Content and Footer are sent in a single file, they are processed as separate parts. It means that Content can be processed successfully, but Footer not successfully. In such case, invoices from Content part remains in Lekuka. In case Content part fails to be processed, Footer part processing is skipped.
- Invoice is created in Lekuka only if it does not yet exist (with the same deviceID, fiscalDayNo, receiptGlobalNo and receiptHash). It means if the same file with the same invoice is sent once again, or the same invoice is sent in another file invoice processing is skipped (because it is already saved in Lekuka).
- If two files have same fiscalDayNo value but different fiscalDayOpened values. First received fiscalDayOpened value according to sequence field is saved, later received value is ignored.

Input parameters:

Name	Type	Mandatory	Description
deviceID	Int	Yes	Device ID.
file	File in multipart/form-data	Yes	File containing invoices and other information related to fiscal period. Base64 encoded.

File:

Name	Type	Mandatory	Description
header	FileHeader	Yes	File header with fiscal period information.
content	FileContent	No*	File content information. Mandatory if invoices are sent.
footer	FileFooter	No*	File footer information. Mandatory if Z report is sent. Fiscal period closure procedure should be initiated.

FileHeader:

Name	Type	Mandatory	Description
deviceID	Int	Yes	Device ID.
fiscalDayNo	Int	Yes	Fiscal period number. Must be same as in Lekuka if fiscal period is opened or greater by 1 if previous fiscal period is closed in case processImmediately is true. Otherwise, fiscal period number must be equal or greater than saved in Lekuka.
fiscalDayOpened	DateTime	Yes	Date and time when fiscal period was opened on a device. Time is provided in local time without time zone information.
fileSequence	Int	Yes	File sequence number in fiscal period files sequence.

FileContent:

Name	Type	Mandatory	Description
receipts	Receipt array	Yes	List of receipts in file. Receipt type description is provided in **Error! Reference source not found.** endpoint description.

FileFooter:

Name	Type	Mandatory	Description
fiscalCounters	FiscalDayCounter array	No	List of counters. Zero value counters must not be submitted to Lekuka. FiscalDayCounter type description provided in **Error! Reference source not found.**closeDay endpoint.
fiscalDayDeviceSignature	SignatureData	Yes	SignatureData structure with SHA256 hash of fiscal period report fields (hash used for signature) and fiscal period report device signature prepared by using device private key as described in section 13.3. **Validation rules:** - fiscalDayDeviceSignature must be valid
receiptCounter	Int	Yes	receiptCounter value of last receipt of current fiscal period.
fiscalDayClosed	DateTime	Yes	Date and time when fiscal period was closed on a device. Time is provided in local time without time zone information.

Output parameters:

Name	Type	Mandatory	Description
operationID	String (60)	Yes	Operation ID assigned by Lekuka.

4.9.1. File example


```
{
	"header": {
		"deviceID": 1111,
		"fiscalDayNo": 5,
		"fiscalDayOpened": "2023-05-30T08:38:54",
		"fileSequence": 2
	},
	"content": {
		"receipts": [
			{
				"receiptType": "FiscalInvoice",
				"receiptCurrency": "LSL",
				"receiptCounter": 5,
				"receiptGlobalNo": 1112,
				"invoiceNo": "IV-2023/1256",
				"receiptDate": "2023-05-30T18:38:54",
				"receiptLinesTaxInclusive": true,
				"receiptLines": [
					{
						"receiptLineType": "Sale",
						"receiptLineSubType": "Product",
						"receiptLineNo": 1,
						"receiptLineHSCode": "85456852",
						"receiptLineName": "Man's shoes",
						"receiptLinePrice": 25,
						"receiptLineQuantity": 1,
						"receiptLineTotal": 25,
						"taxCode": "A",
						"taxRate": 15,
						"taxID": 1
					}
				],
				"receiptTaxes": [
					{
						"taxCode": "A",
						"taxRate": 15,
						"taxID": 1,
						"taxType": "VAT",
						"taxAmount": 3.75,
						"salesAmountWithTax": 28.75
					}
				],
				"receiptPayments": [
					{
						"moneyTypeCode": "Cash",
						"paymentAmount": 28.75
					}
				],
				"receiptTotal": 28.75,
				"receiptPrintForm": "Receipt48",
				"receiptDeviceSignature": {
					"hash": "Yjkjy =",
					"signature": "Yy ="
				}
			},
			{
				//invoice No 2 data
				"receiptType": "FiscalInvoice",
				"receiptCurrency": "LSL"
			},
			{
				//invoice No 3 data
				"receiptType": "FiscalInvoice",
				"receiptCurrency": "LSL"
			}
		]
	},
	"footer": {
		"fiscalDayCounters": [
			{
				"fiscalCounterType": "SaleByTax",
				"fiscalCounterCurrency": "LSL",
				"fiscalCounterTaxRate": 15,
				"fiscalCounterTaxID": 0,
```

```
                    "fiscalCounterMoneyType": "Cash",
                    "fiscalCounterValue": 28.75
                }
            ],
            "fiscalDayDeviceSignature": {
                "hash": "Yjkjy =",
                "signature": "Yy ="
            },
            "receiptCounter": 1,
            "fiscalDayClosed": "2023-05-30T22:38:54"
        }
}
```


Valid sample:


```
{
  "header": {
    "deviceId": 1111,
    "fiscalDayNo": 6,
    "fiscalDayOpened": "2023-05-30T08:38:54",
    "fileSequence": 2
  },
  "content": {
    "receipts": [
      {
        "receiptType": "fiscalInvoice",
        "receiptCurrency": "LSL",
        "receiptCounter": 5,
        "receiptGlobalNo": 1112,
        "invoiceNo": "IV-2023/1256",
        "receiptDate": "2023-05-30T18:38:54",
        "receiptLinesTaxInclusive": true,
        "receiptLines": [
          {
            "receiptLineType": "sale",
            "receiptLineSubType": "product",
            "receiptLineNo": 1,
            "receiptLineHSCode": "85456852",
            "receiptLineName": "Man's shoes",
            "receiptLinePrice": 25,
            "receiptLineQuantity": 1,
            "receiptLineTotal": 25,
            "taxCode": "A",
            "taxRate": 15,
            "taxID": 1
          }
        ],
        "receiptTaxes": [
          {
            "taxCode": "A",
            "taxRate": 15,
            "taxID": 1,
            "taxType": "VAT",
            "taxAmount": 3.75,
            "salesAmountWithTax": 28.75
          }
        ],
        "receiptPayments": [
          {
            "moneyTypeCode": "cash",
            "paymentAmount": 28.75
          }
        ],
        "receiptTotal": 28.75,
        "receiptPrintForm": "receipt48",
        "receiptDeviceSignature": {
          "hash": "bGFiYXM=",
          "signature": "bGFiYXM="
        },
       "receiptType": "fiscalInvoice",
        "receiptCurrency": "LSL"
      }
    ]
  },
  "footer": {
    "fiscalDayCounters": [
      {
        "fiscalCounterType": "saleByTax",
```

```
            "fiscalCounterCurrency": "LSL",
            "fiscalCounterTaxRate": 15,
            "fiscalCounterTaxID": 0,
            "fiscalCounterMoneyType": "cash",
            "fiscalCounterValue": 28.75
          }
        ],
        "fiscalDayDeviceSignature": {
          "hash": "bGFiYXM=",
          "signature": "bGFiYXM="
        },
        "receiptCounter": 1,
        "fiscalDayClosed": "2023-05-30T22:38:54"
    }
}
```


4.10. getFileStatus

getFileStatus (/Device/v1/{deviceID}/SubmittedFileList) endpoint is used by device to get previously sent file processing status from Lekuka.

Request can be sent if DeviceOperatingMode is Offline.

Request will not be accepted, error will be returned to device (as specified in 8.2 Error codes), in these cases:

- device status is other than "*Active*";
- device operating mode is other than *"Offline";*
- request structure is not valid.

Input parameters:

Name	Type	Mandatory	Description
deviceID	Int	Yes	Device ID.
operationID	String (60)	No	Unique operation identifier received in submitFile response. Mandatory if fiscalDayNo is not sent.
fileUploadedFrom	Date	Yes	Date from when uploaded files are needed. Interval between from and to can be no more than 100 days.
fileUploadedTill	Date	Yes	Date till when uploaded files are needed. Interval between from and to can be no more than 100 days.

Output parameters:

Name	Type	Mandatory	Description
operationID	String (60)	Yes	Operation ID assigned by Lekuka.
fileStatus	FileStatus array	Yes	List of files statuses.

FileStatus:

Name	Type	Mandatory	Description
operationID	Int	Yes	Operation ID received in 4.9submitFile request.
fileUploadDate	DateTime	Yes	File placement date and time. Time is provided in local time without time zone information.
deviceId	Int	Yes	Device id.
fileName	String (100)	Yes	File name.
fileProcessingDate	DateTime	No	Date and time when file processing is finished. Returned if fileProcessingStatus is FileProcessingIsSuccessful or FileProcessingWithErrors. Time is provided in local time without time zone information.
fileProcessingStatus	FileProcessingStatus	Yes	File processing status. Possible statuses are defined in section 5.4.11 FileProcessingStatus.
fileProcessingErrorCode	FileProcessingError array	No	List of error codes which appear during submitted file processing, returned if FileProcessingStatus is

FileProcessingWithErrors. Possible codes are defined in section 5.4.12 FileProcessingError

Messages which are shown in case of error during submitted file processing. Possible values:

Enum value	Enum order	Description
IncorrectFileFormat	0	File structure is incorrect, elements sequence is incorrect.
FileSentForClosedDay	1	Fiscal period for which file is sent is closed.
BadCertificateSignature	2	Closing day is not allowed. Bad certificate signature is used.
MissingReceipts	3	Closing day is not allowed. There are missing receipts in fiscal period (“Grey” validation error).
ReceiptsWithValidationErrors	4	Closing day is not allowed. There are receipts with validation errors in fiscal period (“Red” validation error).
CountersMismatch	5	Closing day is not allowed. There are mismatches between counters.
FileExceededAllowedWaitingTime	6	Previous file in a sequence is missing, and maximum allowed waiting time for previous file to be received has exceeded.

4.10.1. TaxType

Type of tax. Possible values:

Enum value	Enum order
Exempt	0
FixedValueLevy	1
NonVAT	2
PercentageLevy	3
VAT	4
	5

.

fiscalDayNo	Int	Yes	Fiscal period number.
fiscalDayOpenedAt	DateTime	Yes	Date and time when fiscal period was opened on a device. Time is provided in local time without time zone information.
fileSequence	Int	Yes	File sequence number in fiscal period files sequence.
ipAddress	String (100)	Yes	Ip address from which file is uploaded.
hasFooter	Boolean	Yes	Information if file had footer
invoicesWithValidationErrors	InvoiceWithValidationError array	No	Invoices with validation errors found by Lekuka. • Returned if there is as least one invoice which has at least one validation error described in: 8.3 Validation errors during specific fiscal day. • In all specific fiscal day files responses retuned all invoices with validations errors submitted with that day.

InvoiceWithValidationError:

Name	Type	Mandatory	Description
receiptCounter	Int	Yes	Daily ascending serial number of receipt assigned by taxpayer's device.
receiptGlobalNo	Int	Yes	Cumulative ascending serial number of total receipts issued since device activation date.
validationErrors	ValidationError array	Yes	Invoice validation errors found by Lekuka. Returned if invoice has at least one validation error described in: 8.3 Validation errors. ValidationError type description is provided in submitReceipt endpoint description.

4.11. closeDay

closeDay (/Device/v2/{deviceID}/CloseDay) endpoint is used to initiate fiscal period closure procedure. This method is allowed when fiscal periods status is "FiscalDayOpened" or "FiscalDayCloseFailed".

Request can't be sent if DeviceOperatingMode is Offline. If DeviceOperatingMode is Offline error DEV01 is received.

In case fiscal period contains at least one "Grey" or "Red" receipt (as specified in 8.3. Validation errors), Lekuka will respond to *closeDay* request with error (fiscal day will remain opened). Otherwise, if fiscal period does not have "Grey" and "Red" receipts, validation of submitted *closeDay* request will be executed. In case of fiscal period validation fails (as specified below in "Validation rules"), fiscal period remains opened, and its status is changed to "FiscalDayCloseFailed".

Input parameters:

Name	Type	Mandatory	Description
deviceID	Int	Yes	Device ID
fiscalDayNo	Int	Yes	Fiscal period number. **Validation rules:** - fiscalDayNo must be the same as provided/received fiscalDayNo value in openDay request.
fiscalDayCounters	FiscalDayCounters array	Yes	List of counters. Zero value counters must not be submitted to Lekuka.
fiscalDayDeviceSignature	SignatureData	Yes	SignatureData structure with SHA256 hash of fiscal period report fields (hash used for signature) and fiscal period report device signature prepared by using device private key as described in section 13.3. **Validation rules:** - fiscalDayDeviceSignature must be valid
receiptCounter	Int	Yes	receiptCounter value of last receipt of current fiscal period.

FiscalDayCounter:

Name	Type	Mandatory	Description
fiscalCounterType	FiscalCounterType	Yes	Counter type.
fiscalCounterCurrency	String (3)	Yes	Counter currency (ISO 4217 currency code). Must be LSL.
fiscalCounterTaxID	Int	No*	Tax ID of counter. Must be provided for all counter types “byTax”.
fiscalCounterTaxRate	Decimal (10,2)	No*	Tax percentage or fixed value of counter. Must be provided for all counter types “byTax”. In case of exempt, this field must not be provided.
fiscalCounterMoneyType	MoneyType	No*	Code of payment mean of counter. Must be provided for counter type “BalanceByMoneyType”.
fiscalCounterValue	Decimal (19,2)	Yes	Counter value in counter currency.

Output parameters:

Name	Type	Mandatory	Description
operationID	String (60)	Yes	Operation ID assigned by Lekuka.

4.12. getServerCertificate

getServerCertificate (/Public/v1/GetServerCertificate) endpoint is used to retrieve Lekuka certificate for Lekuka signature validation.

This API endpoint does not require certificate for authentication.

Input parameters:

Name	Type	Mandatory	Description
thumbprint	Binary (20)	No	Thumbprint of Lekuka signing certificate which should be returned. If field is not provided, currently active Lekuka signing certificate is returned. Together with the certificate, all certificate chain is returned.

Output parameters:

Name	Type	Mandatory	Description
certificate	String array	Yes	Lekuka certificate chain (according to x.509 standard) to validate Lekuka signatures.
certificateValidTill	Date	Yes	Date till when Lekuka signing certificate is valid (despite that in the certificate parameter all the certificate chain is returned, this field shows validity time of the child certificate in the chain). Value is provided in UTC time.

4.13. ping

ping (/Device/v1/{deviceID}/Ping) endpoint is used to report device is online to Lekuka. When device is turned on, it must regularly report to Lekuka that it is online. Reporting periodicity is specified in DeviceReportingFrequencyInMinutes parameter received in response from Lekuka.

Input parameters:

Name	Type	Mandatory	Description
deviceID	Int	Yes	Device ID

Output parameters:

Name	Type	Mandatory	Description
operationID	String (60)	Yes	Operation ID assigned by Lekuka.
reportingFrequency	int	Yes	Reporting frequency in minutes.

5. DATA TYPES

5.1. Address

Address object is used to define address object for returning information from Lekuka and for accepting buyers information.

Name	Type	Mandatory	Description
district	String (100)	Yes	District name
city	String (100)	Yes	City, town, growth point, farming area, mining area
street	String (100)	Yes	Street, stand number, village
houseNo	String (250)	Yes	House number, description of house

5.2. Contacts

Contacts object is used to define Taxpayers and device contact information.

Name	Type	Mandatory	Description
phoneNo	String (20)	No*	Phone number
email	String (100)	No*	E-mail address

* at least one of field is mandatory.

5.3. SignatureData

SignatureData:

Name	Type	Mandatory	Description
hash	Binary (32)	Yes	SHA-256 hash.
signature	Binary (256)	Yes	Cryptographic signature, for which *hash* was used. More details see in “13 Signatures generation and verification rules”. Field length is variable depends on cryptographic algorithm.

SignatureDataEx:

Name	Type	Mandatory	Description
SignatureData fields			
certificateThumbprint	Binary (20)	Yes	SHA-1 Thumbprint of Certificate used for *signature*.

5.4. Enums

Enum, short for "enumerated," is a data type that consists of predefined values. A variable defined as an Enum can store one of the values listed in the Enum declaration.

5.4.1. DeviceOperatingMode

Specifies what are allowed receipt processing modes for this taxpayer, possible values:

Enum value	Enum order
Online	0
Offline	1

5.4.2. FiscalDayStatus

Device fiscal period status, possible values:

Enum value	Enum order
FiscalDayClosed	0
FiscalDayOpened	1
FiscalDayCloseInitiated	2
FiscalDayCloseFailed	3

5.4.3. FiscalDayReconciliationMode

Defines how fiscal period was closed, possible values:

Enum value	Enum order
Auto	0
Manual	1

5.4.4. FiscalCounterType

Counter type, possible values:

Enum value	Enum order
SaleByTax	0
SaleTaxByTax	1
CreditNoteByTax	2
CreditNoteTaxByTax	3
DebitNoteByTax	4
DebitNoteTaxByTax	5
BalanceByMoneyType	6
PayoutByTax	7
PayoutTaxByTax	8

5.4.5. MoneyType

Code of payment mean of counter, possible values:

Enum value	Enum order
Cash	0
Card	1
MobileWallet	2
Coupon	3
Credit	4
BankTransfer	5
Other	6

5.4.6. ReceiptType

Type of receipt. Possible values:

Enum value	Enum order
FiscalInvoice	0
CreditNote	1
DebitNote	2
Receipt	3
Payout	4

5.4.7. ReceiptLineType

Type of receipt line. Possible values:

Enum value	Enum order
Sale	0
Discount	1
Payout	2

5.4.8. ReceiptLineSubType

Subtype of receipt line. Possible values:

Enum value	Enum order
Product	0
Service	1

5.4.9. ReceiptPrintForm

Type of receipt or invoice visual representation template in which form it was printed and delivered to buyer. Possible values:

Enum value	Enum order	Description
Receipt48	0	Printed as receipt on receipt paper, 48 characters per line.
InvoiceA4	1	Printed on A4 paper as invoice.

5.4.10. FiscalDayProcessingError

Messages which are shown in case of error during fiscal period closure. Validations are performed in this sequence. Possible values:

Enum value	Enum order	Description
BadCertificateSignature	0	Closing day is not allowed. Bad certificate signature is used.
MissingReceipts	1	Closing day is not allowed. There are missing receipts in fiscal period (“Grey” validation error).
ReceiptsWithValidationErrors	2	Closing day is not allowed. There are receipts with validation errors in fiscal period (“Red” validation error).
CountersMismatch	3	Closing day is not allowed. There are mismatches between counters.

5.4.11. FileProcessingStatus

File processing status, possible values:

Enum value	Enum order
FileProcessingInProgress	0
FileProcessingIsSuccessful	1
FileProcessingWithErrors	2
WaitingForPreviousFile	3

5.4.12. FileProcessingError

Messages which are shown in case of error during submitted file processing. Possible values:

Enum value	Enum order	Description
IncorrectFileFormat	0	File structure is incorrect, elements sequence is incorrect.
FileSentForClosedDay	1	Fiscal period for which file is sent is closed.

BadCertificateSignature	2	Closing day is not allowed. Bad certificate signature is used.
MissingReceipts	3	Closing day is not allowed. There are missing receipts in fiscal period (“Grey” validation error).
ReceiptsWithValidationErrors	4	Closing day is not allowed. There are receipts with validation errors in fiscal period (“Red” validation error).
CountersMismatch	5	Closing day is not allowed. There are mismatches between counters.
FileExceededAllowedWaitingTime	6	Previous file in a sequence is missing, and maximum allowed waiting time for previous file to be received has exceeded.

5.4.13. TaxType

Type of tax. Possible values:

Enum value	Enum order
Exempt	0
FixedValueLevy	1
NonVAT	2
PercentageLevy	3
VAT	4
	5

5.4.14. TaxRoundingType

Type of tax. Possible values:

Enum value	Enum order
PerReceipt	0
PerReceiptLine	1

6. COUNTERS

With each submitted receipt (FiscalInvoice, Receipt, Payout, CreditNote and DebitNote), counters are updated.

After device finishes a fiscal period, it must close it by sending fiscal period report with counters provided in the table below to E-Invoicing Gateway API. Counter is optional to be sent in case it's value is zero.

Counters list and calculation rules for different types of receipt and different types of receipt lines are provided below. Please take a note to use correct sign when calculating a counter (add or subtract value from counter). In case negative receipt total amount is sent, counters become a negative sign too.

Fiscal period counters are reset after fiscal period close. Starting from a new fiscal period, counters start to be counted from zero.

Table below lists counters and specifies either they are calculated for different tax percents, currencies and/or payment methods:

Counter	By tax	By currency	By payment method	Description
SaleByTax	X	X		Total sales amount (after discount) by tax and by currency during fiscal period including tax. Does not include debit notes, credit notes and payouts.
SaleTaxByTax	X	X		Total tax amount by tax and by currency from sales (after discount) during fiscal period. Does not include debit notes, credit notes and payouts.
CreditNoteByTax	X	X		Total credit notes amount by tax and by currency during fiscal period including tax.
CreditNoteTaxByTax	X	X		Total tax amount by tax and by currency from credit notes during fiscal period.
DebitNoteByTax	X	X		Total debit notes amount by tax and by currency during fiscal period including tax.
DebitNoteTaxByTax	X	X		Total tax amount by tax and by currency from debit notes during fiscal period.
BalanceByMoneyType		X	X	Total collected or paid amount of money, by money type and by currency during fiscal period.
PayoutByTax	X	X		Total sales amount by tax and by currency where document type is Payout during fiscal period including tax.
PayoutTaxByTax	X	X		Total tax amount by tax and by currency from sales where document type is Payout during fiscal period.

Table below shows, which counters each submitted receipt type is changing and which fields from submitted receipt are used:

Receipt type / Counter	Invoice / Receipt	Credit note	Debit note	Payout
SaleByTax	+salesAmountWithTax			
SaleTaxByTax	+taxAmount			
CreditNoteByTax		+salesAmountWithTax		
CreditNoteTaxByTax		+taxAmount		
DebitNoteByTax			+salesAmountWithTax	
DebitNoteTaxByTax			+taxAmount	
BalanceByMoneyType	+paymentAmount	+paymentAmount	+paymentAmount	+paymentAmount
PayoutByTax				+salesAmountWithTax
PayoutTaxByTax				+taxAmount

Note that all values indicated in table above are added to the counter value. If the field is positive – it will increase counter value, if field is negative – it will decrease counter value. So, for example, credit note document which is crediting FiscalInvoice will decrease BalanceByMoneyType counter value as paymentAmount value in such a document will be negative.

7. INTEGRATION SETUP REQUIREMENTS

7.1. Communication and security protocols

E-Invoicing Gateway API can be accessed using HTTPS protocol only. All E-Invoicing Gateway API methods except registerDevice and getServerCertificate use client authentication certificate which is issued by Lekuka.

7.2. Environment addresses

E-Invoicing Gateway API is accessible in testing and production (real) environments. URL to access API:

Environment	URL
Testing environment	https://lekukaapi.rsl.org.ls:8443 Testing environment's API can also be accessed using Swagger on https://lekukaapi.rsl.org.ls:8443/swagger
Production environment	https://lekukaapi.rsl.org.ls

7.3. Authentication and authorization

E-Invoicing Gateway uses mutual TLS authentication (https://en.wikipedia.org/wiki/Mutual_authentication) to authenticate device using device certificate. Device certificate is validated against issuing certificate to allow or deny access to API endpoints.

Note: endpoints 4.1 verifyTaxpayerInformation, 4.2 registerDevice, 4.12 getServerCertificate are public and do not require authentication. After authentication, provided device certificate is checked against issued certificate (see 4.2 registerDevice and 4.3 issueCertificate and methods for device certificate issuing) to check if the device certificate was issued to calling device (by method parameter deviceId) and the device certificate was not revoked.

The E-Invoicing Gateway will return HTTP 401 unauthorized code if:

- The provided device certificate was issued not by E-Invoicing Gateway.
- The provided device certificate was revoked.
- The provided device certificate expired.
- The provided device certificate was not issued to calling device.

7.4. Timeout Settings

E-Invoicing Gateway API response timeout for any synchronous operation – 30 seconds.

8. ERRORS

In case of API error, the system will return 4xx or 5xx http error code with response body containing a detailed problem details structure as described in https://www.rfc-editor.org/rfc/rfc7807 .

ProblemDetails:

Name	Type	Mandatory	Description
type	String	Yes	
title	String	Yes	human readable problem definition
status	Integer	Yes	Http status code
errorCode	String	No	specific error code, for exact meaning see below

8.1. Http statuses

API can return such http statuses for errors:

Http status	Description
400	bad request - the message is malformed and could not be processed by E-Invoicing Backend Gateway
401	Authentication error (see Authentication and authorization)
404	Resource not found (call to not existing endpoint)
405	method not allowed - trying to access API using unsupported HTTP methos, e.g., POST to get config
422	Unprocessable Content - the instructions given by device to E-Invoicing Backend Gateway are incorrect, the response object ProblemDetails should contain ErrorCode to indicate the exact failing condition (e.g., DEV01 - device is blocked and therefore no instructions could be processed from such device)
500	Infrastructure error - the E-Invoicing Backend Gateway server is not available, or some infrastructure error occurred. The device should retry to send message later.
502	Bad gateway - the E-Invoicing Backend Gateway server could not be contacted. The device should retry to send message later.

8.2. Error codes

Error Code	Description	API methods
DEV01	Device not found or not active	All
DEV02	Activation key is incorrect	registerDevice verifyTaxpayerInformation
DEV03	Certificate request is invalid	registerDevice issueCertificate
DEV04	Device model is blacklisted	All
DEV05	Taxpayer is not active	All
DEV06	Device model and version is not registered in Lekuka	All
RCPT01	Submitting receipt is not allowed. Fiscal period is closed, or fiscal period close initiated	submitReceipt
RCPT02	Submit receipt failed. The receipt structure invalid or field requirements not satisfied (e.g., provided field value length is greater then allowed)	submitReceipt

FISC01	Open day is not allowed	openDay
FISC03	Closing day is not allowed. Close day is in progress.	closeDay
FISC04	Closing day is not allowed. Fiscal period not opened.	closeDay
FILE01	File is too big. Allowed file size: 3 MB.	submitFile
FILE02	File structure invalid or field requirements not satisfied (e.g not sent mandatory fields).	submitFile getFileStatus
FILE03	Device operating mode is not Offline.	submitFile getFileStatus
FILE04	File sent for already closed day or closing in progress.	submitFile
FILE05	Device id in input parameters and file header are different	submitFile

8.3. Validation errors

When *sumbitReceipt* request is received by Lekuka and it is not rejected, Lekuka validates receipt data. In case previous receipt is missing, validation rules which require previous receipt to be present are temporarily marked as grey and are revalidated when previous receipt is received. After a validation, receipt is stored and signed by Lekuka. Receipt validation status may be valid or invalid. In case of invalid receipt, validation errors are categorized by one of these colors:

Color	Description
Grey	This means that received receipt violates receipt chain and makes a gap in receipt chain (previous receipt is missing). Such receipt is marked in “Grey”. With each of the next received receipts, such “Grey” receipt will be revalidated (in case newly received receipt is the previous for the “Grey” receipt). After repeated validation it will remain “Grey” (if newly received receipt is not the previous for it) or become valid. Fiscal period will not be allowed to be closed automatically if it has at least one “Grey” receipt.
Yellow	“Yellow” validation errors are minor ones. Fiscal period will be allowed to be closed automatically if it contains only “Yellow” validation errors.
Red	“Red” validation errors are major ones. Fiscal period fill is not allowed to be closed automatically if it has at least one “Red” receipt.

Possible validation errors and their color codes:

Validation error	Color	Do validation requires previous receipt to be present?	Validation error text
RCPT010	Red	No	Wrong currency code is used
RCPT011	Red	Yes	Receipt counter is not sequential
RCPT012	Red	Yes	Receipt global number is not sequential
RCPT013	Red	No	Invoice number is not unique
RCPT014	Yellow	No	Receipt date is earlier than fiscal period opening date
RCPT015	Red	No	Credited/debited invoice data is not provided
RCPT016	Red	No	No receipt lines provided
RCPT017	Red	No	Taxes information is not provided
RCPT018	Red	No	Payment information is not provided
RCPT019	Red	No	Invoice total amount is not equal to sum of all invoice lines

RCPT020	Red	No	Invoice signature is not valid
RCPT021	Red	No	VAT tax is used in invoice while taxpayer is not VAT taxpayer
RCPT022	Red	No	Invoice sales line price and / or line total must be greater than 0 (less than 0 for Credit note), discount line price and / or line total must be less than 0 for Invoice and Receipt, payout line price and / or line total must be less than 0 (greater than 0 for Debit note)
RCPT023	Red	No	Invoice line quantity, must be positive
RCPT024	Red	No	Invoice line total is not equal to unit price * quantity
RCPT025	Red	No	Invalid tax is used
RCPT026	Red	No	Incorrectly calculated tax amount
RCPT027	Red	No	Incorrectly calculated total sales amount (including tax)
RCPT028	Red	No	Payment amount must be greater than or equal to 0 (less than or equal to 0 for Credit note which is created from Receipt or Fiscal invoice, Debit note created from Payout or Payout)
RCPT029	Red	No	Credited/debited invoice information provided for regular invoice
RCPT030	Red	Yes	Invoice date is earlier than previously submitted receipt date
RCPT031	Yellow	No	Invoice is submitted with the future date
RCPT032	Red	No	Credit/debit note refers to non-existing invoice
RCPT033	Red	No	Credited/debited invoice is issued more than 12 months ago
RCPT034	Red	No	Note for credit/debit note is not provided
RCPT035	Red	No	Total credit note amount exceeds original invoice amount
RCPT036	Red	No	Credit/debit note uses other taxes than are used in the original invoice
RCPT037	Red	No	Invoice total amount is not equal to sum of all invoice lines and taxes applied
RCPT038	Red	No	Invoice total amount is not equal to sum of sales amount including tax in tax table
RCPT039	Red	No	Invoice total amount is not equal to sum of all payment amounts
RCPT040	Red	No	Invoice total amount must be greater than or equal to 0 (less than or equal to 0 for Credit note which is created from Receipt or Fiscal invoice, Debit note created from Payout or Payout document)
RCPT041	Yellow	No	Invoice is issued after fiscal period end
RCPT042	Red	No	Credit/debit note uses other currency than is used in the original invoice
RCPT043	Red	No	Mandatory buyer data fields are not provided
RCPT044	Red	No	Receipt line type payout only allowed in payout document or debit/credit note document if credit or debit note is created for payout document
RCPT045	Red	No	Receipt line types discount, sale are not allowed in payout document and debit/credit note document if credit or debit note is created for payout document
RCPT046	Red	No	Buyer data is not provided
RCPT049	Red	No	Tax table is missing some taxes indicated in invoice lines
RCPT050	Red	No	Wrongly indicated taxes in invoice lines
RCPT051	Red	No	Mandatory applied for quantity field is not provided
RCPT052	Red	No	Receipt counter is duplicated
RCPT053	Red	No	Receipt global No is duplicated
RCPT054	Red	No	Buyer TIN cannot be the same as seller TIN

9. REQUIREMENTS FOR DEVICES

This chapter specifies requirements for devices to be met:

1. Device must open a new fiscal period before issuing receipts and invoices.
2. Device if internet connection is available must retrieve configuration from Lekuka (*getConfig* endpoint) before opening a new fiscal period.
3. Device must save data from *getConfig* response about taxpayer and/or its branch (taxpayer name, address, contacts, etc.) and use it for receipt and invoice printing.
4. Device must track the time passed from opening a fiscal period, that it would not exceed maximum allowed fiscal period length (specified in parameter taxPayerDayMaxHrs) and forbid issuing new receipts and invoices after that.
5. Device must inform user about the approaching fiscal period end. Notification must be shown to the user a few hours before maximum fiscal period length is reached. The exact number of hours left to maximum fiscal period length is specified in parameter taxpayerDayEndNotificationHrs.
6. Device must assign receiptGlobalNo value to a receipt in a sequential order starting from 1 and continue numbering despite fiscal period close.
7. In case receiptGlobalNo value becomes very big and taxpayer would like to reset it, this can be done by submitting the first receipt in a new fiscal period.
8. Device must assign receiptCounter value to a receipt in a sequential order starting from 1 and continue numbering only in the same fiscal period. After fiscal period closure, receiptCounter must be reset to 0.
9. Device must not allow to add goods or services with VAT tax to receipt or invoice if taxpayer is not a VAT taxpayer (*VATNumber* value is received in *getConfig* response). In case taxpayer gets VAT number in the middle of fiscal period, device must not allow to issue new receipts or invoices and must require closing fiscal period first.
10. Fiscal period opening message must be sent immediately after opening a fiscal period, however if there is no connection, it can be delayed. Actual for devices which operating mode is Online.
11. Receipt must be sent to E-Invoicing Gateway API only after successfully opening a fiscal period.
12. Device must send receipt to E-Invoicing Gateway API only after finishing it (when receipt is printed). Actual for devices which operating mode is Online.
13. Device must send receipt to E-Invoicing Gateway API one by one in ascending receiptGlobalNo order, not skipping any of receipt. In case submission of receipt failed, issue must be fixed, and receipt submission must be repeated. Actual for devices which operating mode is Online.

14. Device must send receipt to E-Invoicing Gateway API immediately after finishing it in case there is an internet connection and there are no waiting receipts to be sent, otherwise receipt must be put to the queue on device and send later when connection will be restored. Actual for devices which operating mode is Online.
15. Device must update counters after issuing a receipt and reset counters after starting a new fiscal period as specified in 5.4.14. TaxRoundingType

Type of tax. Possible values:

Enum value	Enum order
PerReceipt	0
PerReceiptLine	1

16. Counters.
17. Device must renew certificate which is near to expiration before its expiration date.
18. If device operating mode is Offline, device must send file content only for a single closed fiscal period. File cannot contain invoices from more than one fiscal period.
19. If file is bigger than 3 MB, content should be split into separate files, footer can’t be split to two or more separate parts.
20. If fiscal period has a lot of receipts, it is recommended to split receipts and Z report to separate files.
21. If device which operating mode is Offline prepared more than one file for a single fiscal period, these files must be sent to Lekuka one by one in sequential order.
22. Device which operating mode is Offline must be able to export file with invoices if user requests it for manual upload to Lekuka Self-service portal.
23. User must not uninstall application from device if there are unsent invoices.

10. STANDARD RECEIPT, INVOICE AND REPORT VIEWS

10.1. Receipt48 view

Receipt48 view is used for tax inclusive invoice printed on receipt paper, which can print 48 characters of text per line.

Tax invoice template:

Receipt	Receipt field name
	[1] Taxpayer logo
Company legal name	[2] Taxpayer company legal name
TIN: 123111000	[3] Taxpayer TIN
VAT No: 12341234	[4] VAT number
Downtown location	[5] Taxpayer's branch name
Government Office Complex, Kingsway Road 12480a, Maseru	[6] Taxpayer's branch address
rsl@email.com	[7] Taxpayer's branch e-mail
(0266)22 758 801	[8] Taxpayer's branch phone
TAX INVOICE	[9] Static text: Tax invoice or Invoice or Credit note or Debit note or Receipt or Payout notice
Buyer	[10] Buyer's information
Company ABC, Ltd.	[11] Buyer's registered name
Food Market ABC	[12] Buyer's trade name
TIN: 19870123	[13] Buyer's TIN
12 Street Maseru	[14] Buyer's address
john.smith@email.com	[15] Buyer's e-mail
(081) 20875	[16] Buyer's phone
Invoice No: 15/451 Fiscal period No: 45	[17], [18], [19] Invoice No (receiptCounter, receiptGlobalNo), Fiscal period number
Customer reference No: CISN-000040321	[20] Customer reference number
Device Serial No: 12345678901234567890	[21] Device serial number
Device ID: 6543210	[22] Device ID
Date: 02/27/2024 14:25	[23] Receipt date and time
Description **Amount**	List of receipt items, if If line type is discount static text "Discount" is shown.
Swings, adult ticket 13200.00 VT	[30] , [33] , [34], Item name, total amount, tax code
Item2 with very long name which does not fit in a single line	
3 each @ 5000.00 15000.00 VT	[31], [32] item quantity, unit price
Item3 name	
0.555 @ 1081.08 600.00 NV	
Item4 name 2400.00 EX	
Discount:Item4 name -1200.00 EX	
Total LSL **30015.00**	[35], [36] receipt currency, total amount to pay
LSL Cash 5000.00	
LSL Card 25015.00	[35], [37], [38] receipt currency, payment method, paid amount
Number of Items **5.555**	[39] number of items
Net Amount 1200.00	Tax table
VAT (Exempt) 0.000	
Gross Amount 1200.000	
	[40], [41], [42], [43], [44] tax code, tax percent, amount without tax, tax amount, total amount inclusive tax
Levy name Attraction percent	Levy table
Rate 10%	
Levy Amount 10.00	
	[40], [54], [53], [57], [43] tax code, levy name, levy rate, levy amount
Levy name Attractions 1 levy	
Rate 5.00	
Levy Amount 5.00	
Invoice is issued after purchasing goods	[49] Credit/debit/invoice note
	[45] QR code
Verification code: 4C8B-E276-6333-0417	[46] Receipt verification code
You can verify this receipt manually at	
https://www.google.com	[48] URL

Credit note template:

Receipt	Receipt field name
	[1] Taxpayer logo
Company legal name	[2] Taxpayer company legal name
TIN: 123111000	[3] Taxpayer TIN
VAT No: 12341234	[4] VAT number
Downtown location	[5] Taxpayer's branch name
Government Office Complex, Kingsway Road 12480a, Maseru	[6] Taxpayer's branch address
rsl@email.com	[7] Taxpayer's branch e-mail
(0266)22 758 801	[8] Taxpayer's branch phone
CREDIT NOTE	[9] Static text: Tax invoice or Invoice or Credit note or Debit note or Receipt or Payout notice
Buyer	[10] Buyer's information
Company ABC, Ltd.	[11] Buyer's registered name
Food Market ABC	[12] Buyer's trade name
TIN: 19870123	[13] Buyer's TIN
12 Street Maseru	[14] Buyer's address
john.smith@email.com	[15] Buyer's e-mail
(081) 20875	[16] Buyer's phone
Invoice No: 15/451 Fiscal period No: 45	[17], [18], [19] Invoice No (receiptCounter, receiptGlobalNo), Fiscal period number
Customer reference No: CISN-000040321	[20] Customer reference no
Device Serial No: 12345678901234567890	[21] Device serial number
Device ID: 6543210	[22] Device ID
Date: 02/02/2024 18:48	[23] Receipt date and time
	Printed ony for Debit and Credit note
CREDITED INVOICE	[24] Static text "Credited invoice" or "Debited invoice"
Device Serial No: 9876543210123456789	[25]Credited/debited invoice device serial No
Invoice No: 450 Date: 03/07/2024 10:10	[26], [27] Credited/debited invoice No (receiptGlobalNo) and date
Customer reference No: CISN-000040320	[28] Credited/debited invoice customer reference No
Description **Amount**	List of receipt items, if If line type is discount static text "Discount" is shown.
Item1 name 13200.00 VT	[30] , [33] , [34], Item name, total amount, tax code
Item2 with very long name which does not fit in a single line	
3 each @ 5000.00 15000.00 VT	[31], [32] item quantity, unit price
Item3 name	
0.555 @ 1081.08 600.00 NV	
Item4 name 2400.00 EX	
Discount:Item4 name -1200.00 EX	
Total LSL **30000.00**	[35], [36] receipt currency, total amount to pay
LSL Cash 5000.00	
LSL Card 25000.00	[35], [37], [38] receipt currency, payment method, paid amount
Number of Items 5.555	[39] number of items
VAT% **Net.Amt** **VAT** **Amount**	Tax table
EX 1200.00 0.00 1200.00	
NV 0 600.00 0.00 600.00	
VT 15 24521.74 3678.26 28200.00	[40], [41], [42], [43], [44] tax code, tax percent, amount without tax, tax amount, total amount inclusive tax
Invoice is issued after purchasing goods	[45] Credit/debit/invoice note
	[46] QR code
Verification code: 4C8B-E276-6333-0417	[47] Receipt verification code
You can verify this receipt manually at https://receipt.rsl.org/	[48] URL

10.2. InvoiceA4 view

InvoiceA4 view is used for tax inclusive, or tax exclusive invoice printed on A4 paper.

Tax inclusive invoice template

Verification code
4C8B-E276-6333-0417 [46]
You can verify this receipt manually at
https://receipt.rsl.org/ [48]

[1] Taxpayer logo

[45] QR code

TAX INVOICE [9]

SELLER
Company legal name [2]
TIN: 1234567890 [3]
VAT No: 12345678 [4]
Downtown location [5]
Government Office Complex, Kingsway Road 258, Maseru [6]
rsl@email.com [7]
(0266) 22 758 891 [8]

BUYER
Company ABC, Ltd. [11]
Food Market ABC [12]
TIN: 19870123 [13]
12 Southgate Hwange [14]
john.smith@email.com [15]
(081) 20875 [16]

Invoice No: 15[17]/451[18]
Customer reference No: CISN-0000040012 [20]
Device Serial No: 12345678901234567890 [21]

Fiscal period No: 45 [19]
Date: 27/02/2024 10:48 [23]
Device ID: 674473 [22]

Code [29]	**Description [30]**	**Qty [31]**	**Price [32]**	**VAT [34.1]**	**Total amount (incl. tax) [33.1]**
12345678	Swings adult ticket	1	13200,00	1721,74	13200,00
11223344	Item2 name with very long name	3	5000,00	1956,52	15000,00
12312312	Item3 name	1	600,00	0,00	600,00
12341234	Item4 name	1	2400,00	-	2400,00
12341234	Discount: Item4 name	1	-1200,00	-	-1200,00

Total15% [41] VAT [56]	3678,26 [44]
Total15% [53] Levy [54]	35,24
Total Attractions 1 Levy [54] (2.50) [57]	35,24
Invoice total, LSL [36]	**33120,00 [35]**

Invoice is issued after purchasing goods according to agreement No555 [49]

Tax inclusive credit note template

[1] Taxpayer logo

Verification code
4C8B-E276-6333-0417 [46]
You can verify this receipt manually at
https://receipt.rsl.org/ [48]

[45] QR code

CREDIT NOTE [9]

SELLER
Company legal name [2]
TIN: 1234567890 [3]
VAT No: 12345678 [4]
Downtown location [5]
Government Office Complex, Kingsway Road 258, Maseru [6]
rsl@email.com [7]
(0266) 22 758 891 [8]

BUYER
Company ABC, Ltd. [11]
Food Market ABC [12]
TIN: 19870123 [13]
12 Southgate Hwange [14]
john.smith@email.com [15]
(081) 20875 [16]

Invoice No: 15[17]/451[18]
Customer reference No: CISN-0000040012 [20]
Device Serial No: 12345678901234567890 [21]

Fiscal period No: 45 [19]
Date: 27/02/2024 14:48 [23]
Device ID: 674473 [22]

CREDITED INVOICE [24]

Invoice No: 450 [26]
Customer reference No: CISN-0000040011 [28]
Device Serial No: 12345678901234567890 [25]

Date: 27/02/2024 10:48 [27]

Code [29]	Description [30]	Qty [31]	Price [32]	VAT [34.1]	Total amount (incl. tax) [33.1]
12345678	Item1 name	1	13200.00	1721.74	13200.00
11223344	Item2 name with very long name which does not fit in a single line	3	5000.00	1956.52	15000.00
12312312	Item3 name	1	600.00	0.00	600.00
12341234	Item4 name	1	2400.00	-	2400.00
12341234	Discount: Item4 name	1	-1200.00	-	-1200,00

Total15% [41]	VAT	3678,26 [44]
	Invoice total, LSL [36]	**33120.00 [35]**

Invoice is issued after purchasing goods according to agreement No555 [49]

Tax exclusive invoice template

Verification code
4C8B-E276-6333-0417 [46]
You can verify this receipt manually at
https://receipt.rsl.org/ [48]

[1] Taxpayer logo

[45] QR code

TAX INVOICE [9]

SELLER
Company legal name [2]
TIN: 1234567890 [3]
VAT No: 12345678 [4]
Downtown location [5]
Government Office Complex, Kingsway Road 258, Maseru [6]
rsl@email.com [7]
(0266) 22 758 891 [8]

BUYER
Company ABC, Ltd. [11]
Food Market ABC [12]
TIN: 19870123 [13]
12 Southgate Hwange [14]
john.smith@email.com [15]
(081) 20875 [16]

Invoice No: 15[17]/451[18]
Customer reference No: CISN-0000040012 [20]
Device Serial No: 12345678901234567890 [21]

Fiscal period No: 45 [19]
Date: 26/02/2024 18:48 [23]
Device ID: 674473 [22]

Code [29]	**Description [30]**	**Qty [31]**	**Price [32]**	**Amount (excl. tax) [50]**	**VAT [34.2]**	**Total amount (incl. tax) [33.2]**
12345678	Item1 name	1	13200.00	13200.00	1980.00	15180.00
11223344	Item2 name with very long name which does not fit in a single line	3	5000.00	15000.00	2250.00	17250.00
12312312	Item3 name	1	600.00	600.00	0.00	600.00
12341234	Item4 name	1	2400.00	2400.00	-	2400.00
12341234	Discount: Item4 name	1	-1200.00	-1200.00	-	-1200.00

Total (excl. tax) [47]	30000.00
Total15% [41] VAT	4320.00 [44]
Invoice total, LSL [36]	**34320.00 [35]**

Invoice is issued after purchasing goods according to agreement No555 [49]

Tax exclusive credit note template

[1] Taxpayer logo

Verification code
4C8B-E276-6333-0417 [46]
You can verify this receipt manually at
https://receipt.rsl.org/ [48]

[45] QR code

CREDIT NOTE [9]

SELLER	BUYER
Company legal name [2]	Company ABC, Ltd. [11]
TIN: 1234567890 [3]	Food Market ABC [12]
VAT No: 12345678 [4]	TIN: 19870123 [13]
Downtown location [5]	12 Southgate Hwange [14]
Government Office Complex, Kingsway Road 258, Maseru [6]	john.smith@email.com [15]
rsl@email.com [7]	(081) 20875 [16]
(0266) 22 758 891 [8]	

Invoice No: 15[17]/451[18]	Fiscal period No: 45 [19]
Customer reference No: CISN-0000040012 [20]	Date: 26/02/2024 18:48 [23]
Device Serial No: 12345678901234567890 [21]	Device ID: 674473 [22]

CREDITED INVOICE [24]

Invoice No: 450 [26]	Date: 26/02/2024 10:48 [27]
Customer reference No: CISN-0000040011 [28]	
Device Serial No: 12345678901234567890 [25]	

Code [29]	**Description [30]**	**Qty [31]**	**Price [32]**	**Amount (excl. tax) [50]**	**VAT [34.2]**	**Amount (excl. tax) [33.2]**
12345678	Item1 name	1	13200.00	13200.00	1980.00	15180.00
11223344	Item2 name with very long name which does not fit in a single line	3	5000.00	15000.00	2250.00	17250.00
12312312	Item3 name	1	600.00	600.00	0.00	600.00
12341234	Item4 name	1	2400.00	2400.00	-	2400.00
12341234	Discount: Item4 name	1	-1200.00	-1200.00	-	-1200,00

Total (excl. tax) [47]	30000.00
Total15% [41] VAT	4320.00 [44]
Invoice total, LSL [36]	**34320.00 [35]**

Invoice is issued after purchasing goods according to agreement No555 [49]

10.3. Receipt and invoice view fields descriptions

Fields in "Field name in which device sends data to LEKUKA" refers to E-Invoicing Gateway API endpoint *submitReceipt* fields. Fields in "Field name in which device receives data from Lekuka" refers to E-Invoicing Gateway API endpoint *getConfig* fields.

Ob. No	Object name	Object description	Field name in which device sends data to Lekuka	Field name in which device receives data from Lekuka	Used in	Mandatory
	Taxpayer block					
[1]	Taxpayer's logo	Taxpayer's logo. If printer is not capable to print logo, field is not displayed.	-	-	Receipt48 InvoiceA4	N
[2]	Taxpayer's name	Taxpayer's company legal name.	-	taxpayerName	Receipt48 InvoiceA4	Y
[3]	Taxpayer's TIN	Taxpayer's identification number, displayed with label "TIN: ".	-	taxpayerTIN	Receipt48 InvoiceA4	Y
[4]	Taxpayer's VAT No	Taxpayer's VAT number. Must be displayed if taxpayer is VAT taxpayer	-	vatNumber	Receipt48 InvoiceA4	Y
	Taxpayer address and contacts block					
[5]	Taxpayer's branch name	Taxpayer's branch name (to which device is assigned) Field displayed only if it differs from Taxpayer's name.	-	deviceBranchName	Receipt48 InvoiceA4	Y*
[6]	Taxpayer's branch address	Taxpayer's branch address, where device is located. Displayed in this order: houseNumber, street, city, district	-	deviceBranchAddress	Receipt48 InvoiceA4	Y
[7]	Taxpayer's branch e-mail	E-mail address.	-	deviceBranchContacts. email	Receipt48 InvoiceA4	N
[8]	Taxpayer's branch phone number	Phone number.	-	deviceBranchContacts. phoneNo	Receipt48 InvoiceA4	N
	Tax invoice block					
[9]	Label	Static text "TAX INVOICE" if document is invoice and taxpayer is VAT payer or "INVOICE" if document is invoice and taxpayer is non VAT payer or "CREDIT NOTE" if document is credit note or "DEBIT NOTE" if document is debit note or "RECEIPT" if document is receipt or "PAYOUT NOTICE" if document is payout	-	-	Receipt48 InvoiceA4	Y
	Buyer block Buyer block is displayed only if buyer information is provided.					
[10]	Block label	Static text "Buyer:".	-	-	Receipt48 InvoiceA4	Y

[11]	Buyer's Name.	Buyer's register name (or individual person name, foreign company name or not registered trader name).	buyerRegisterName	-	Receipt48 InvoiceA4	Y
[12]	Buyer's trade name	Buyer's trade name	buyerTradeName	-	Receipt48 InvoiceA4	N
[13]	Buyer TIN	Buyer's TIN.	buyerTIN	-	Receipt48 InvoiceA4	N
[14]	Buyer's address	Buyer's address. Displayed in this order: houseNumber, street, city, district	buyerAddress	-	Receipt48 InvoiceA4	N
[15]	Buyer's e-mail	Buyer's e-mail address.	buyerContacts. email	-	Receipt48 InvoiceA4	N
[16]	Buyer's phone number	Purchaser's phone number. Field displayed if not empty.	buyerContacts. phoneNo	-	Receipt48 InvoiceA4	N
	Receipt information block					
[17]	Invoice No	Receipt number in fiscal period - current day receipt counter.	receiptCounter	-	Receipt48 InvoiceA4	Y
[18]	Invoice No	Receipt global No.	receiptGlobalNo	-	Receipt48 InvoiceA4	Y
[19]	Fiscal period No	Receipt fiscal period number.	-	fiscalDayNo	Receipt48 InvoiceA4	Y
[20]	Customer reference No	Customer reference number must be unique at taxpayer level	invoiceNo		Receipt48 InvoiceA4	Y
[21]	Device Serial No	Device serial number.	-	deviceSerialNo	Receipt48 InvoiceA4	Y
[22]	Device ID	Device ID (assigned by LEKUKA during device registration).	deviceID	-	Receipt48 InvoiceA4	Y
[23]	Receipt date and time	Receipt date and time.	receiptDate	-	Receipt48 InvoiceA4	Y
	Credit note or Debit Note information block (displayed only in case of receipt type Credit Note or Debit Note)					
[24]	Label	Static text "Credited invoice" or "Debited invoice". Displayed only for Credit notes and Debit notes respectively.	-	-	Receipt48 InvoiceA4	Y
[25]	Device Serial No	Device Serial No of credited or debited receipt	creditNote.deviceID	-	Receipt48 InvoiceA4	Y
[26]	Invoice No	Invoice No (receiptGlobalNo) of credited or debited receipt.	creditNote. receiptGlobalNo	-	Receipt48 InvoiceA4	Y
[27]	Receipt date	Date of credited or debited receipt	receiptDate of credited or debited receipt	-	Receipt48 InvoiceA4	Y
[28]	Customer reference No	Customer reference No of credited or debited	invoiceNo of credited or debited receipt	-	Receipt48 InvoiceA4	Y

		receipts, is unique at taxpayer level				
	Receipt lines block					
[29]	Item code	Receipt line item code.	receiptLineHSCode	-	InvoiceA4	N
[30]	Item name	Item name. • If item name does not fit into 1 row it is split into more rows depending on item name length. • If ReceiptLineType is Discount static text "Discount:" is added as prefix.	receiptLineName	-	Receipt48 InvoiceA4	Y
[31]	Quantity	Receipt line item Quantity. May not be displayed in case quantity is 1 if it is Receipt48.	receiptLineQuantity,	-	Receipt48 InvoiceA4	N
[32]	Unit price	Receipt line item unit price. May not be displayed in case quantity is 1 1 if it is Receipt48. If Quantity is more than 1 word "each" is shown in Receipt48 view. In InvoiceA4 tax inclusive template - price is shown including tax. In InvoiceA4 tax exclusive template - price is shown excluding tax.	receiptLinePrice	-	Receipt48 InvoiceA4	N
[33]	Total amount	Receipt line total.	receiptLineTotal	-	Receipt48 InvoiceA4	Y
[33.1]	Total amount (incl. tax)	Receipt line total	receiptLineTotal	-	Receipt48 InvoiceA4	Y
[33.2]	Total amount (incl. tax)	Receipt line total inclusive tax Amount = receiptLineTotal * (100+taxRate)	-	-	Receipt48 InvoiceA4	Y
[34]	Tax code	Tax code of receipt line.	taxCode	-	Receipt48	N
[34.1]	VAT	VAT amount of receipt line. VAT = receiptLineTotal * taxRate/(100+taxRate). If tax doesn't have percent, is null, shown minus sign.	-	-	Receipt48 InvoiceA4	Y
[34.2]	VAT	VAT amount of receipt line. VAT = receiptLineTotal * taxRate. If tax doesn't have percent, is null, shown minus sign.	-	-	Receipt48 InvoiceA4	Y
[50]	Total amount (excl. tax)	Receipt line total amount excluding tax (receiptLineTotal value for tax exclusive invoice)	receiptLineTotal	-	InvoiceA4	Y
~~[51]~~	~~Levy name~~	~~If for receipt line additionally is applied FixedValueLevy shown applied levy name~~	~~-~~	~~taxName~~	~~Receipt48 InvoiceA4~~	~~N~~
[52]	Levy amount	If for receipt line additionally is applied FixedValueLevy shown applied levy amount. Levy amount = appliedForQuantity * taxRate	-	-	Receipt48 InvoiceA4	N

		if appliedForQuantity is sent or Levy amount = receiptLineQuantity * taxRate if appliedForQuantity is not sent				
~~[55]~~	~~Levy quantity~~	~~If for receipt line additionally is applied FixedValueLevy shown for which quantity levy is applied:~~ ~~appliedForQuantity if appliedForQuantity is sent~~ ~~receiptLineQuantity if appliedForQuantity is not sent~~	~~receiptLineQuantity or appliedForQuantity~~	~~-~~	~~InvoiceA4~~	~~N~~
	Receipt settlement block					
[36]	Currency	Receipt currency code.	receiptCurrency	-	Receipt48 InvoiceA4	Y
[35]	Total receipt amount	Total receipt amount to be paid.	receiptTotal	-	Receipt48 InvoiceA4	Y
[37]	Payment method	Payment method.	moneyTypeCode	-	Receipt48	Y
[38]	Total paid	Total paid by payment method.	paymentAmount	-	Receipt48 InvoiceA4	Y
	Number of Items block					
[39]	Number of Items	Number of items in receipt. SUM(Quantity) of all Sales lines. If item is weighed weight is added. For example: if one item is quantitative and quantity is 4, and another item is weighed, and weight is 0,555 kg, number of items should be 4+0,555=4,555. Always positive number (for credit note Number of Items also will be showed as positive number).	-	-	Receipt48	Y
	Taxes block Receipt rows are grouped by tax rate and tax code (taxes are ordered by type in this order: VAT, Exempt, NonVAT, WithholdingTax, PercentageLevy, FixedValueLevy, then by taxId in InvoiceA4 view, taxes are ordered by type in this order: VAT, Exempt, NonVAT, WithholdingTax then by taxId in Receipt48 view). This block is not shown on receipt if taxpayer is not VAT taxpayer.					
[40]	Tax code	Tax code	-	-	Receipt48	N
[41]	Vat %	Tax percentage. In case of exempt, no value is displayed.	-	-	Receipt48 InvoiceA4	Y
[42]	Net.Amt	Amount without tax, salesAmountWithTax - taxAmount	-	-	Receipt48 InvoiceA4	Y
[43]	VAT	Tax amount	taxAmount	-	Receipt48 InvoiceA4	Y
[44]	Amount	Sales amount with tax	salesAmountWithTax	-	Receipt48 InvoiceA4	Y
[47]	Total (excl. tax)	Used in InvoiceA4 when exclusive tax line totals are used. It sums total	-	-	InvoiceA4	N

		amount exclusive tax column.				
[53]	Levy percentage	If invoice has applied PercentageLevy shown levy percentage value	-	taxRate	InvoiceA4	N
[54]	Levy name	If invoice has applied FixedValueLevy, PercentageLevy shown levy name	-	taxName	InvoiceA4	N
[56]	Tax type	Shown tax type name if tax type is VAT or NonVAT	-	taxType	InvoiceA4	N
[57]	Levy rate	If invoice has applied FixedValueLevy shown levy fixed value rate	-	taxRate	InvoiceA4	N
	Levies block Receipt rows are grouped by levy rate and levy code (levies are ordered by type in this order: PercentageLevy, FixedValueLevy, then by taxId). This block is not shown on receipt if no additional levies were sent.					
[40]	Tax code	Levy tax code	-	-	Receipt48	N
[54]	Levy name	If receipt has applied FixedValueLevy, PercentageLevy shown levy name, shown first 20 symbols	-	taxName	Receipt48	N
[53]	Levy percentage	If receipt has applied PercentageLevy shown levy percentage value + static text "%"	-	taxRate	Receipt48	N
[57]	Levy rate	If receipt has applied FixedValueLevy shown levy fixed value rate	-	taxRate	Receipt48	N
[43]	Levy amount	Levy amount	taxAmount	-	Receipt48	N
	Receipt verification block					
[45]	QR code	Generated QR code. More details in *Receipt QR code rules*. Optional if printer cannot print QR picture.	-	-	Receipt48 InvoiceA4	N*
[46]	Receipt verification code	QR code value (receiptQRData) of generate QR displayed in four characters groups separated by "- ". More details in *Receipt QR code rules*.	-	-	Receipt48 InvoiceA4	Y
[48]	URL	URL for QR validation.	-	qrUrl	Receipt48 InvoiceA4	Y
[49]	Note	Note for credit note/debit note/invoice.	receiptNotes	-	Receipt48 InvoiceA4	Y*

10.4. Z Report

Receipt			Receipt field name
Company legal name			[1] Taxpayer company legal name
TIN: 123111000			[2] Taxpayer TIN
VAT No: 12341234			[3] VAT number
Downtown location			[4] Taxpayer's branch name
Government Office Complex, Kingsway Road 12480a, Maseru			[5] Taxpayer's branch address
rsl@email.com			[6] Taxpayer's branch e-mail
(0266) 22 758 891			[7] Taxpayer's branch phone
--			
Z REPORT			[8] Static text: Z REPORT
--			
Fiscal period No: 45			[9] Fiscal period number
Fiscal period opened: 03/02/2024 18:01			[10] Fiscal period start date
Fiscal period closed: 04/02/2024 18:01			[11] Fiscal period end date
Device Serial No: 98765432101234567 89			[12]Device serial No
Device Id: 1450			[13]Device Id
--			
Daily totals			[14] Static text: Daily totals
--			List of counters in particular currency. Currency/counters information is shown only if there are sales with corresponding currency.
LSL			[15] Currency (currencies are in alphabetical order)
--			
Total net sales			[16] Static text: Total net sales
Net,VAT 15%		10,000.00	[17], [18] Static text "Net", tax name, net amount (sales without tax) (ordered by tax percentage descending)
Net,VAT 9%		5,000.00	
Net,Non-VAT 0%		15,000.00	
Net,Exempt		25,000.00	
Total net amount		**55,000.00**	[19], [20] Static text: Total net amount, total amount
--			
Total taxes			[21] Static text: Total taxes
Tax,VAT 15%		1,500.00	[22] [23] Static text "Tax", tax name, tax amount (tax amount) (ordered by tax percentage descending)
Tax,VAT 9%		450.00	
Total tax amount		**1,950.00**	[24], [25] Static text: Total tax amount, total amount
--			
Total levies			[52] Static text: Total levies
Tax,Levy 15%		200.00	[22] [23] Static text "Tax", tax name, tax amount (tax amount) (ordered by type in this order: Percentage levy, Amount levy, then by taxId)
Tax,Attractions 1 levy		100.00	[53], [54] Static text: Total levies amount, total levies amount
Total levies amount		**300.00**	
--			
Total gross sales			[26] Static text: Total gross sales
Total,VAT 15%		11,500.00	[27], [28] Static text "Total", tax name, total amount (sales with tax) (ordered by tax percentage descending)
Total,VAT 9%		5,450.00	
Total,Non-VAT 0%		15,000.00	
Total,Exempt		25,000.00	
Total gross amount		**57,250.50**	[29], [30] Static text: Total gross amount, total amount
--			
Documents	**Quantity**	**Total amount**	[31], [32],[33] Static texts: Documents,Quantity, Total amount
Invoices	50	56,350.50	[34] ,[35],[36] Static text "Invoices", quantity of documents, total amount (sales with tax for invoices)
Receipts	3	900.00	[37], [38],[39] Static text "Receipts", quantity of documents, total amount (sales with tax for receipts)
Credit notes	1	-300.00	[40] ,[41],[42] Static text "Credit notes", quantity of documents, total amount (sales with tax for credit notes, value can be negative)
Debit notes	1	500.00	[43], [44], [45] Static text "Debit notes", quantity of documents, total amount (sales with tax for debit notes)
Payouts	2	-200.00	[46], [47], [48] Static text "Payouts", quantity of documents, total amount (sales with tax for payouts)
Total documents	**55**	**57,250.50**	[49], [50], [51] Static text "Total documents", total quantity, total sum of amounts
==			

10.5. Z report fields description

Z report are daily reports. Fields are described according confirmCertificate
confirmCertificate (/Device/v2/{deviceID}/ConfirmCertificate) endpoint is used to confirm that new certificate is received.
It is mandatory to confirm after each certificate renewal.

Input parameters:

Name	Type	Mandatory	Description
deviceID	Int	Yes	Device ID
thumbprint	Binary (20)	Yes	Thumbprint of Lekuka signing certificate which is confirmed.

Output parameters:

Name	Type	Mandatory	Description
operationID	String (60)	Yes	Operation ID assigned by Lekuka.

getConfig, openDay, getStatus APIs, Enums and TaxRoundingType
Type of tax. Possible values:

Enum value	Enum order
PerReceipt	0
PerReceiptLine	1

Counters.

Ob. No	Object name	Object description	Field name
Taxpayer block			
[1]	Taxpayer name	Taxpayer's company legal name.	taxpayerName
[2]	Taxpayer TIN	Taxpayer's identification number, displayed with label "TIN: ".	taxpayerTIN
[3]	Taxpayer VAT No	Taxpayer's VAT number, displayed with label "VAT No:". Must be displayed if taxpayer is VAT taxpayer	vatNumber
Taxpayer address and contacts block			
[4]	Taxpayer's branch name	Taxpayer's branch name (to which device is assigned) Field displayed only if it differs from Taxpayer's name.	deviceBranchName
[5]	Taxpayer's branch address	Taxpayer's branch address, where device is located. Displayed in this order: houseNumber, street, city, province	deviceBranchAddress
[6]	Taxpayer's branch e-mail	E-mail address.	deviceBranchContacts. email
[7]	Taxpayer's branch phone number	Phone number.	deviceBranchContacts. phoneNo
Report block			
[8]	Label	Static text "Z REPORT"	-
[9]	Fiscal period No	Fiscal period number.	fiscalDayNo
[10]	Fiscal period opening date	Fiscal period opening date.	fiscalDayOpened
[11]	Fiscal period closing date	Fiscal period closing date. Is shown if FiscalDayStatus is FiscalDayClosed.	fiscalDayClosed

[12]	Device Serial No	Device serial number.	deviceSerialNo
[13]	Device ID	Device ID (assigned by LEKUKA during device registration).	deviceID
Daily totals block			
[14]	Label	Static text: Daily totals.	-
[15]	Currency	List of counters in particular currency. Block for particular currency is shown only if there are sales with corresponding currency. Blocks by currency is listed in alphabetical ascending order by currency code.	fiscalCounterCurrency
[16]	Total net sales text	Static text “Total net sales”.	-
[17]	Net name	List of Net amounts for each tax. Static text “Net” + tax name. Taxes are ordered by tax percentage in descending order. If tax percent is equal to 0%, or exempt – line for that tax is not shown.	taxName
[18]	Net amount	Total net amount (sales without tax) by tax. NB. Credit note counter is added total because that it’s value is negative.	SaleByTax – SaleTaxByTax + CreditNoteByTax – CreditNoteTaxByTax + DebitNoteByTax – DebitNoteTaxByTax by particular tax.
[19]	Total net amount text	Static text “Total net amount”	-
[20]	Total net amount	Total net amount for all values, all taxes. NB. Credit note counter is added total because that it’s value is negative.	Sum of (SaleByTax – SaleTaxByTax + CreditNoteByTax – CreditNoteTaxByTax + DebitNoteByTax - DebitNoteTaxByTax)
[21]	Total taxes text	Static text “Total taxes”. This block is skipped if no taxes were collected (there were no sales with tax which percent is greater than 0%)	-
[22]	Tax name	List of Tax amounts for each tax where tax type is VAT, Exempt. Static text “Tax” + tax name. Taxes are ordered by tax percentage in descending order. If tax percent is equal to 0%, or exempt – line for that tax is not shown. In levies case levies are ordered by type in this order: PercentageLevy, FixedValueLevy, then by tax id	taxName
[23]	Tax amount	Tax amount where tax type is VAT, Exempt. NB. Credit note counter is added total because that it’s value is negative.	SaleTaxByTax + CreditNoteTaxByTax + DebitNoteTaxByTax by particular tax where tax type is VAT, Exempt.
[24]	Total tax amount text	Static text “Total tax amount”	-
[25]	Total tax amount	Total tax amount for all values, all taxes where tax type is VAT, Exempt. NB. Credit note counter is added total because that it’s value is negative.	Sum of (SaleTaxByTax + CreditNoteTaxByTax + DebitNoteTaxByTax) where tax type is VAT, Exempt.
[52]	Total levies text	Static text “Total levies”.	-

[53]	Total levies amount text	Static text "Total levies amount".	-
[54]	Levies amount	Total tax amount for all values, all taxes where tax type is PercentageLevy, FixedValueLevy. NB. Credit note counter is added total because that it's value is negative.	Sum of (SaleTaxByTax + CreditNoteTaxByTax + DebitNoteTaxByTax) where tax type is PercentageLevy, FixedValueLevy.
[26]	Total gross sales text	Static text "Total gross sales".	-
[27]	Tax name	List of Total gross amounts for each tax. Static text "Total" + tax name. Taxes are ordered by tax percentage in descending order. If tax percent is equal to 0, or exempt - line for that tax is not shown.	taxName
[28]	Gross sales amount	Gross sales amount (sales with tax). NB. Credit note counter is added total because that it's value is negative.	SaleByTax + CreditNoteByTax + DebitNoteByTax by particular tax.
[29]	Total gross amount text	Static text "Total gross amount"	-
[30]	Total gross amount	Total gross amount for all values, all taxes. NB. Credit note counter is added total because that it's value is negative.	Sum of (SaleByTax + CreditNoteByTax + DebitNoteByTax)
[31]	Documents text	Static text "Documents".	-
[32]	Quantity text	Static text "Quantity".	-
[33]	Total amount text	Static text "Total amount".	-
[34]	Invoices text	Static text "Invoices".	-
[35]	Quantity of invoices	Quantity of invoices during fiscal period.	Number of issued documents where ReceiptType=FiscalInvoice
[36]	Total amount for invoices	Total amount (sales with tax for invoices).	Sum of SaleByTax
[37]	Receipts text	Static text "Receipts".	-
[38]	Quantity of receipts	Quantity of receipts during fiscal period.	Number of issued documents where ReceiptType=Receipt
[39]	Total amount for receipts	Total amount (sales with tax for receipts).	Sum of SaleByTax
[40]	Credit notes text	Static text "Credit notes".	-
[41]	Quantity of credit notes	Quantity of credit notes during fiscal period.	Number of issued documents where ReceiptType=CreditNote
[42]	Total amount for credit notes	Total amount (sales with tax for credit notes). Is shown with minus sign.	Sum of CreditNoteByTax
[43]	Debit notes text	Static text "Debit notes".	-
[44]	Quantity of debit notes	Quantity of debit notes during fiscal period.	Number of issued documents where ReceiptType=DebitNote
[45]	Total amount for debit notes	Total amount (sales with tax for debit notes).	Sum of DebitNoteByTax
[46]	Payouts text	Static text "Payouts".	-
[47]	Quantity of payouts	Quantity of payouts during fiscal period.	Number of issued documents where ReceiptType=Payout
[48]	Total amount for payouts	Total amount (sales with tax for payouts).	Sum of PayoutByTax
[49]	Total documents text	Static text "Total documents".	-
[50]	Total quantity	Quantity of documents during fiscal period. NB. Credit note counter is added total because that it's value is negative.	Number of issued documents

[51]	Total amount	Total amount for all documents, all taxes. NB. Credit note counter is added total because that it’s value is negative.	Sum of SaleByTax + Sum of CreditNoteByTax + Sum of DebitNoteByTax

11. RECEIPT QR CODE RULES

Each issued receipt and invoice must contain QR code value printed as text and preferably also QR code picture (in case printer is capable to print it). QR code represents deep link URL with receipt identification information. QR code consists of this information:

Name	Length*	Description
qrUrl		URL for QR validation, received in getConfig, qrUrl field.
deviceID	10	Device ID represented in 10 digits number with leading zeros.
receiptDate	8	Invoice date (receiptDate field value) represented in 8 digits (format: ddMMyyyy).
receiptGlobalNo	10	Receipt global number (receiptGlobalNo field value) issued by device represented in 10 digits with leading zeros
receiptQrData	16	Receipt QR data field (first 16 hexadecimal characters of MD5 hash from ReceiptDeviceSignature value).

* - length specifies a fixed length of value, which should be included in QR. In case value is shorter than indicated length, leading zeros must be added in front.

Example:

Name	Example No 1	Example No 2
qrUrl	https://invoice.rsl.org.ls/	https:// invoice.rsl.org.ls/
deviceID	0000000321	0000000322
receiptDate	03042023	04042023
receiptGlobalNo	1112223331	0000001332
receiptQrData	5C8BE27623330417	F10B0478B3B14678
Result (receiptQrCode)	https:// invoice.rsl.org.ls/00000003210304 202311122233315C8BE27623330417	https:// invoice.rsl.org.ls/00000003220404 20230000001332F10B0478B3B14678
QR		

12. CERTIFICATE SIGNING REQUEST (CSR) AND CERTIFICATE EXAMPLES

12.1. Example keys used

NOTE.
Those example keys are supplied only for illustration of CSR and Certificate generation, they are in unencrypted form and should **NEVER BE USED IN REAL LIFE**.
Device should generate their own keys and securely store them in encrypted form, never letting private key to go outside of device.

12.1.1. ECC ECDSA on SECG secp256r1

ECC ECDSA on SECG secp256r1 (ANSI prime256v1, NIST P-256) private key used in examples:


```
-----BEGIN EC PRIVATE KEY-----
MHcCAQEEIBXgREh8BvsXj0FjjcZ29EQiVjWGJuqHQp55+LlZd6waoAoGCCqGSM49
AwEHoUQDQgAE+79w72O6UYOJc9mfO8EjMEl9uysJaWJ0kVellIj46atl7FAG4NpY
VDe6t5pTSWlM6qCj5qKealESKalMnV32qQ==
-----END EC PRIVATE KEY-----
```


This key in textual representation form:

```
Private-Key: (256 bit)
priv:
    15:e0:44:48:7c:06:fb:17:8f:41:63:8d:c6:76:f4:
    44:22:56:35:86:26:ea:87:42:9e:79:f8:b9:59:77:
    ac:1a
pub:
    04:fb:bf:70:ef:63:ba:51:83:89:73:d9:9f:3b:c1:
    23:30:49:7d:bb:2b:09:69:62:74:91:57:a5:94:88:
    f8:e9:ab:65:ec:50:06:e0:da:58:54:37:ba:b7:9a:
    53:49:69:4c:ea:a0:a3:e6:a2:9e:6a:51:12:29:a9:
    4c:9d:5d:f6:a9
ASN1 OID: prime256v1
NIST CURVE: P-256
```

12.1.2. RSA 2048

RSA 2048 private key used in examples:


```
-----BEGIN RSA PRIVATE KEY-----
MIIEpQIBAAKCAQEA52kCd2bXK+W72vC0+KlQ+tLUBNBsYNk9Gg+NPTxfD+92lFg9
sPW8B1MFxTO+Kpw5MRzArB6M3LZ3pj0O525vLcgT301bridwTgpSfzqtoHFhgTox
My94lMiYSK94w2RxslaMyDm4dCXGfU5AlAiuuegVFz056jV/Ik7jFjQLG5GQnRhW
tGkI2TKZLOlBsjhxUKLKPwUCPtkfZmDig/fXE1XigqSOGMoxQ9BTxJ/i8wWU3AR1
Tjou41EFisaOKpan4xqeRnvNrOs0eIccBFRXBCB78tLTZgE9yqgwoQ5oQpNm845O
kFXQ0F770cy0sx+CO2OnpdQisRK8WDx95kJkIQIDAQABAoIBAFar6/KQoBKe7ucn
tIBV2jC3ehV7grwbYVk7bej7jZdIVx9klWaMAyqzG7wqjxUiggE1BazxnEymQtYO
lGB16ko5X8gJD0eBGf0AvLlOXu1yydQ+2WKUaxM+tlqy7gYwvqzO4de0VrOZ2mfg
QSuwvNCAbjXQBrsD4mQVK9SLFYXzHuYPXDVgTnBktjVQpedV+gqdLVo8yMm1Fpv3
bB/tvotBmxHKsVR6AZmGME1ttO0HR4L6Ha4d1ao5FK4+LtvowFSpSssQ7wyczJmO
ualE4qt/S87Fvm1hYU7VLbm7pAcZMa15FhIVcWZar2RRwwhvm7XwZbnippNgETMC
9bDsogECgYEA87D38++I3eCVm+bJ8Qpsu8yKVS2Bqv+dAyFLG9wFmA6OJhNrpQ+f
JxSmv6sxiz492j4I+uHXuo9ylSg8NQmdtouNRD2Gb0Xb5K6W0zftnoj05P8zJkkR
Z9RyoSXppLblazzT6LqUSk1/PWsX3OXlliNEh4babqsU6fartlWrqtECgYEA8xk9
yz46DOKtu5GEl7i2pe2mMXNG3Ed/Bas1mwa1isnKPFjNNGLuMaITYvXDK+/O9hqO
ESWIRg/sjQ1NaHrbZ3izYgqBPyPVX8inA/w+4Yb6CVHY+9KJGVqzqrvlWwkMkd6K
IqL4SQAxWJQ1Ua1znbuB5TZ6tcZ4QcbVVcgB2FECgYEAwK+xn2RLqIUoRvmZu8ou
Z+A3kVpGKVusXwk4RnMWyUDZDSpV91H+2fvuTaejqSIx7hsXJqjk11MNmvsRgC52
UhzOOqMbZWirkoqqH6Eddjl8yoUvgJpN9Pd7HAjKUb98b+rM9DxzfL0CWyIO4E+3
1ZtVWIQ8uzzzcHvnEmlzL8ECgYEA5lPOFpl4yuijDwqLBG3AsGoAgu3j/6XGFgrn
```

mWC79SnH8XF5y97ILEKR97s/Fov6HXd/j4NuIGPKDsLByvJMmzbjT0sAtmAvNLea
ds4yjeAjW10vJzmNKHalsGiioKRsQnEFlFewwwnptzGFa0PaPWKBajk5/qxzGG9Z
hhMgnGECgYEAh3K9PsQSTT2tjxuOyyYjlDqDn7exl81i8XLeVKDOL7uQM6m3VICo
F8KdKY2hrwe71pQMI6+P+GkDgx4J7qWO1EMZdMDcUDgzQemUQiUuFDFZ7FnUOkUS
2RbcYyo9m5kGzHzfQPAAlk9J1tJ3/xm8Hi44b2ZpiCpHYNW+RtB2qjA=
-----END RSA PRIVATE KEY-----

This key in textual representation form:

```
RSA Private-Key: (2048 bit, 2 primes)
modulus:
    00:e7:69:02:77:66:d7:2b:e5:bb:da:f0:b4:f8:a9:
    50:fa:d2:d4:04:d0:6c:60:d9:3d:1a:0f:8d:3d:3c:
    5f:0f:ef:76:94:58:3d:b0:f5:bc:07:53:05:c5:33:
    be:2a:9c:39:31:1c:c0:ac:1e:8c:dc:b6:77:a6:3d:
    0e:e7:6e:6f:2d:c8:13:df:4d:5b:ae:27:70:4e:0a:
    52:7f:3a:ad:a0:71:61:81:3a:31:33:2f:78:94:c8:
    98:48:af:78:c3:64:71:b2:56:8c:c8:39:b8:74:25:
    c6:7d:4e:40:94:08:ae:b9:e8:15:17:3d:39:ea:35:
    7f:22:4e:e3:16:34:0b:1b:91:90:9d:18:56:b4:69:
    08:d9:32:99:2c:e9:41:b2:38:71:50:a2:ca:3f:05:
    02:3e:d9:1f:66:60:e2:83:f7:d7:13:55:e2:82:a4:
    8e:18:ca:31:43:d0:53:c4:9f:e2:f3:05:94:dc:04:
    75:4e:3a:2e:e3:51:05:8a:c6:8e:2a:96:a7:e3:1a:
    9e:46:7b:cd:ac:eb:34:78:87:1c:04:54:57:04:20:
    7b:f2:d2:d3:66:01:3d:ca:a8:30:a1:0e:68:42:93:
    66:f3:8e:4e:90:55:d0:d0:5e:fb:d1:cc:b4:b3:1f:
    82:3b:63:a7:a5:d4:22:b1:12:bc:58:3c:7d:e6:42:
    64:21
publicExponent: 65537 (0x10001)
privateExponent:
    56:ab:eb:f2:90:a0:12:9e:ee:e7:27:b4:80:55:da:
    30:b7:7a:15:7b:82:bc:1b:61:59:3b:6d:e8:fb:8d:
    97:48:57:1f:64:95:66:8c:03:2a:b3:1b:bc:2a:8f:
    15:22:82:01:35:05:ac:f1:9c:4c:a6:42:d6:0e:94:
    60:75:ea:4a:39:5f:c8:09:0f:47:81:19:fd:00:bc:
    b9:4e:5e:ed:72:c9:d4:3e:d9:62:94:6b:13:3e:b6:
    5a:b2:ee:06:30:be:ac:ce:e1:d7:b4:56:b3:99:da:
    67:e0:41:2b:b0:bc:d0:80:6e:35:d0:06:bb:03:e2:
    64:15:2b:d4:8b:15:85:f3:1e:e6:0f:5c:35:60:4e:
    70:64:b6:35:50:a5:e7:55:fa:0a:9d:2d:5a:3c:c8:
    c9:b5:16:9b:f7:6c:1f:ed:be:8b:41:9b:11:ca:b1:
    54:7a:01:99:86:30:4d:6d:b4:ed:07:47:82:fa:1d:
    ae:1d:d5:aa:39:14:ae:3e:2e:db:e8:c0:54:a9:4a:
    cb:10:ef:0c:9c:cc:99:8e:b9:a9:44:e2:ab:7f:4b:
    ce:c5:be:6d:61:61:4e:d5:2d:b9:bb:a4:07:19:31:
    ad:79:16:12:15:71:66:5a:af:64:51:c3:08:6f:9b:
    b5:f0:65:b9:e2:a6:93:60:11:33:02:f5:b0:ec:a2:
    01
prime1:
    00:f3:b0:f7:f3:ef:88:dd:e0:95:9b:e6:c9:f1:0a:
    6c:bb:cc:8a:55:2d:81:aa:ff:9d:03:21:4b:1b:dc:
    05:98:0e:8e:26:13:6b:a5:0f:9f:27:14:a6:bf:ab:
    31:8b:3e:3d:da:3e:08:fa:e1:d7:ba:8f:72:95:28:
    3c:35:09:9d:b6:8b:8d:44:3d:86:6f:45:db:e4:ae:
    96:d3:37:ed:9e:88:f4:e4:ff:33:26:49:11:67:d4:
    72:a1:25:e9:a4:b6:e5:6b:3c:d3:e8:ba:94:4a:4d:
    7f:3d:6b:17:dc:e5:e5:96:23:44:87:86:da:6e:ab:
    14:e9:f6:ab:b6:55:ab:aa:d1
prime2:
    00:f3:19:3d:cb:3e:3a:0c:e2:ad:bb:91:84:97:b8:
```

```
        b6:a5:ed:a6:31:73:46:dc:47:7f:05:ab:35:9b:06:
        b5:8a:c9:ca:3c:58:cd:34:62:ee:31:a2:13:62:f5:
        c3:2b:ef:ce:f6:1a:8e:11:25:88:46:0f:ec:8d:0d:
        4d:68:7a:db:67:78:b3:62:0a:81:3f:23:d5:5f:c8:
        a7:03:fc:3e:e1:86:fa:09:51:d8:fb:d2:89:19:5a:
        b3:aa:bb:e5:5b:09:0c:91:de:8a:22:a2:f8:49:00:
        31:58:94:35:51:ad:73:9d:bb:81:e5:36:7a:b5:c6:
        78:41:c6:d5:55:c8:01:d8:51
    exponent1:
        00:c0:af:b1:9f:64:4b:a8:85:28:46:f9:99:bb:ca:
        2e:67:e0:37:91:5a:46:29:5b:ac:5f:09:38:46:73:
        16:c9:40:d9:0d:2a:55:f7:51:fe:d9:fb:ee:4d:a7:
        a3:a9:22:31:ee:1b:17:26:a8:e4:d7:53:0d:9a:fb:
        11:80:2e:76:52:1c:ce:3a:a3:1b:65:68:ab:92:8a:
        aa:1f:a1:1d:76:39:7c:ca:85:2f:80:9a:4d:f4:f7:
        7b:1c:08:ca:51:bf:7c:6f:ea:cc:f4:3c:73:7c:bd:
        02:5b:22:0e:e0:4f:b7:d5:9b:55:58:84:3c:bb:3c:
        f3:70:7b:e7:12:69:73:2f:c1
    exponent2:
        00:e6:53:ce:16:99:78:ca:e8:a3:0f:0a:8b:04:6d:
        c0:b0:6a:00:82:ed:e3:ff:a5:c6:16:0a:e7:99:60:
        bb:f5:29:c7:f1:71:79:cb:de:c8:2c:42:91:f7:bb:
        3f:16:8b:fa:1d:77:7f:8f:83:6e:20:63:ca:0e:c2:
        c1:ca:f2:4c:9b:36:e3:4f:4b:00:b6:60:2f:34:b7:
        9a:76:ce:32:8d:e0:23:5b:5d:2f:27:39:8d:28:76:
        a5:b0:68:a2:a0:a4:6c:42:71:05:94:57:b0:c3:09:
        e9:b7:31:85:6b:43:da:3d:62:81:6a:39:39:fe:ac:
        73:18:6f:59:86:13:20:9c:61
    coefficient:
        00:87:72:bd:3e:c4:12:4d:3d:ad:8f:1b:8e:cb:26:
        23:94:3a:83:9f:b7:b1:97:cd:62:f1:72:de:54:a0:
        ce:2f:bb:90:33:a9:b7:54:80:a8:17:c2:9d:29:8d:
        a1:af:07:bb:d6:94:0c:23:af:8f:f8:69:03:83:1e:
        09:ee:a5:8e:d4:43:19:74:c0:dc:50:38:33:41:e9:
        94:42:25:2e:14:31:59:ec:59:d4:3a:45:12:d9:16:
        dc:63:2a:3d:9b:99:06:cc:7c:df:40:f0:00:96:4f:
        49:d6:d2:77:ff:19:bc:1e:2e:38:6f:66:69:88:2a:
        47:60:d5:be:46:d0:76:aa:30
```

12.1.3. Sample Root CA certificate

Sample Root CA certificate:


```
-----BEGIN CERTIFICATE-----
MIIFdDCCA1ygAwIBAgIUSEjdSuepCe1JZA+JcdootQDDP1kwDQYJKoZIhvcNAQEL
BQAwUTELMAkGA1UEBhMCTFMxITAfBgNVBAoMGFJldmVudWUgU2VydmljZXMgTGVz
b3RobzEfMB0GA1UEAwwWUlNMICoqU0FNUExFKiogUm9vdCBDQTAgFw0yNDA0MDIx
NDQxMTZaGA8yMTI0MDMwOTE0NDExNlowUTELMAkGA1UEBhMCTFMxITAfBgNVBAoM
GFJldmVudWUgU2VydmljZXMgTGVzb3RobzEfMB0GA1UEAwwWUlNMICoqU0FNUExF
KiogUm9vdCBDQTCCAiIwDQYJKoZIhvcNAQEBBQADggIPADCCAgoCggIBALdMYUfX
hyz9VreTgT5pvsl4MDMdHsKxagn1BJu7eJ7bBC3SwVMY6vueGQoDKB58KMYd7v/+
GgafPYyZAtvfJkWOUBBYfc+ncrujGM/YhoEqmrvwcJ+wTY75IjIGRWmEisZ/QZGY
hYJqZRpjnfwP0cjHvaWbAb5cYfe7Mb0CwaFc4L5tdADe5WTVNe6XB+ynedU6tQHX
TnP311+s0h+TFFJCMZafosd1UMkrPQmLuKk+DS1MKj0Z8Kg+RtC2RQXIy3dJjN31
G9SjgY+MnXHqOkIBgzC7nXlBQrH2qZs88iMhHg0NGA9hvaRhx15VhnIKVpCvJxIj
qi9QrYwPpXvRn1Kbf7y9Og8AS6u/xRv9lAgH2bwWaPuDfgmPukGWKTZ7KukWJ9Aj
n62cObMG2yPwSpM9uhrTWooX83sTG9SBoZ83F6AcV8v0mDLQ9HbrFugznzz2CV6i
/pNYsEaZZ1GaU/IKmT+s5mXaRWXkPHKPAOolfIS1JJ2EJ/PYMBdPtSioflviHSjT
KoQDSX44X7J8ZxeKHhlXPhfwY6HEZEeZSzV3Fis+3iURz5QOiwaS5uaHYmht9NgT
zP95CGcPJcMokRX1JN76y21k7l3WL9VZ+Yu66+9xLwbHh7Nx8QhSDdNhmfKoSjG7
GLppY3ffN5wePZRPL3EhX9ag/hkxnvmQnolRAgMBAAGjQjBAMB0GA1UdDgQWBBSN
```

```
IPNR32QLqv+sYVOnTrnKhxSDoDAPBgNVHRMBAf8EBTADAQH/MA4GA1UdDwEB/wQE
AwIBhjANBgkqhkiG9w0BAQsFAAOCAgEAihBA1KxQqUcoZ6pOhf0FqMcNdr6tSewS
S+H0vELq8ViGcPkyVqDUqeqjFTPfA+Szx/aS4wUQ7U72uqxoRoUVD2aimtP/UvhL
NVdGPcQUb40srKtw7sw04l/BnQWgkMYAHyVf/B8uSXTYFHkIRhdTCjjDxaaHf0FY
8ZIMkUdq290rAD0fAXOypAjRr4DqFonPzk/8k4T/w9z5nDaIi0W5ANi6Y4pv1tNC
iJh8BO926YbtqmO3TORl1HcD0A5k77g3wjtkhB9gHJYkvwf+lZUqMzPPfeuhuSfh
nbqNi1w8mzBHlPXzABuLHCiVCDtzIlKUZ0TqOR0bHbW3CRw0ar651eyIlZjJpn9t
pNwqUzmJ+7yqgR5CR+RTYKZYviXgwOoWGxCTsMcqt2j8O3Mt9rX4duyOe/rvObcu
tHjrBebmw3WV0GzhfXUhaF+XAQ8pFH/SNe+nXfkNxJ741ugf7ppwbr9Tc2cr+556
C4K60jgLgaBmuluDrZGqK7kY4O7hBXfLWSLhDmIukzA1U5TGw4QbijNZhxMOmRRa
nmmkaiNxpWWI+NwTU8nZYToH/qs99U1kyjXdOZgMCrlteF5BWY120tyI4s17Cybj
oMJvpVZga+gmrCh9y9l2Od/51aUoOD8wQAVhTQEd7VMPxP3P9fePw5QxKc0vy+r4
DDmJgS4sMhY=
-----END CERTIFICATE-----
```


Textual representation information:

```
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            48:48:dd:4a:e7:a9:09:ed:49:64:0f:89:71:da:28:b5:00:c3:3f:59
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=LS, O=Revenue Services Lesotho, CN=RSL **SAMPLE** Root CA
        Validity
            Not Before: Apr  2 14:41:16 2024 GMT
            Not After : Mar  9 14:41:16 2124 GMT
        Subject: C=LS, O=Revenue Services Lesotho, CN=RSL **SAMPLE** Root CA
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (4096 bit)
                Modulus:
                    00:b7:4c:61:47:d7:87:2c:fd:56:b7:93:81:3e:69:
                    be:c9:78:30:33:1d:1e:c2:b1:6a:09:f5:04:9b:bb:
                    78:9e:db:04:2d:d2:c1:53:18:ea:fb:9e:19:0a:03:
                    28:1e:7c:28:c6:1d:ee:ff:fe:1a:06:9f:3d:8c:99:
                    02:db:df:26:45:8e:50:10:58:7d:cf:a7:72:bb:a3:
                    18:cf:d8:86:81:2a:9a:bb:f0:70:9f:b0:4d:8e:f9:
                    22:32:06:45:69:84:8a:c6:7f:41:91:98:85:82:6a:
                    65:1a:63:9d:fc:0f:d1:c8:c7:bd:a5:9b:01:be:5c:
                    61:f7:bb:31:bd:02:c1:a1:5c:e0:be:6d:74:00:de:
                    e5:64:d5:35:ee:97:07:ec:a7:79:d5:3a:b5:01:d7:
                    4e:73:f7:d7:5f:ac:d2:1f:93:14:52:42:31:96:9f:
                    a2:c7:75:50:c9:2b:3d:09:8b:b8:a9:3e:0d:2d:4c:
                    2a:3d:19:f0:a8:3e:46:d0:b6:45:05:c8:cb:77:49:
                    8c:dd:f5:1b:d4:a3:81:8f:8c:9d:71:ea:3a:42:01:
                    83:30:bb:9d:79:41:42:b1:f6:a9:9b:3c:f2:23:21:
                    1e:0d:0d:18:0f:61:bd:a4:61:c7:5e:55:86:72:0a:
                    56:90:af:27:12:23:aa:2f:50:ad:8c:0f:a5:7b:d1:
                    9f:52:9b:7f:bc:bd:3a:0f:00:4b:ab:bf:c5:1b:fd:
                    94:08:07:d9:bc:16:68:fb:83:7e:09:8f:ba:41:96:
                    29:36:7b:2a:e9:16:27:d0:23:9f:ad:9c:39:b3:06:
                    db:23:f0:4a:93:3d:ba:1a:d3:5a:8a:17:f3:7b:13:
                    1b:d4:81:a1:9f:37:17:a0:1c:57:cb:f4:98:32:d0:
                    f4:76:eb:16:e8:33:9f:3c:f6:09:5e:a2:fe:93:58:
                    b0:46:99:67:51:9a:53:f2:0a:99:3f:ac:e6:65:da:
                    45:65:e4:3c:72:8f:00:ea:25:7c:84:b5:24:9d:84:
                    27:f3:d8:30:17:4f:b5:28:a8:7e:5b:e2:1d:28:d3:
                    2a:84:03:49:7e:38:5f:b2:7c:67:17:8a:1e:19:57:
                    3e:17:f0:63:a1:c4:64:47:99:4b:35:77:16:2b:3e:
                    de:25:11:cf:94:0e:8b:06:92:e6:e6:87:62:68:6d:
```

```
                    f4:d8:13:cc:ff:79:08:67:0f:25:c3:28:91:15:f5:
                    24:de:fa:cb:6d:64:ee:5d:d6:2f:d5:59:f9:8b:ba:
                    eb:ef:71:2f:06:c7:87:b3:71:f1:08:52:0d:d3:61:
                    99:f2:a8:4a:31:bb:18:ba:69:63:77:df:37:9c:1e:
                    3d:94:4f:2f:71:21:5f:d6:a0:fe:19:31:9e:f9:90:
                    9e:89:51
                Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Subject Key Identifier:
                8D:20:F3:51:DF:64:0B:AA:FF:AC:61:53:A7:4E:B9:CA:87:14:83:A0
            X509v3 Basic Constraints: critical
                CA:TRUE
            X509v3 Key Usage: critical
                Digital Signature, Certificate Sign, CRL Sign
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:
        8a:10:40:d4:ac:50:a9:47:28:67:aa:4e:85:fd:05:a8:c7:0d:
        76:be:ad:49:ec:12:4b:e1:f4:bc:42:ea:f1:58:86:70:f9:32:
        56:a0:d4:a9:ea:a3:15:33:df:03:e4:b3:c7:f6:92:e3:05:10:
        ed:4e:f6:ba:ac:68:46:85:15:0f:66:a2:9a:d3:ff:52:f8:4b:
        35:57:46:3d:c4:14:6f:8d:2c:ac:ab:70:ee:cc:34:e2:5f:c1:
        9d:05:a0:90:c6:00:1f:25:5f:fc:1f:2e:49:74:d8:14:79:08:
        46:17:53:0a:38:c3:c5:a6:87:7f:41:58:f1:92:0c:91:47:6a:
        db:dd:2b:00:3d:1f:01:73:b2:a4:08:d1:af:80:ea:16:89:cf:
        ce:4f:fc:93:84:ff:c3:dc:f9:9c:36:88:8b:45:b9:00:d8:ba:
        63:8a:6f:d6:d3:42:88:98:7c:04:ef:76:e9:86:ed:aa:63:b7:
        4c:e4:65:d4:77:03:d0:0e:64:ef:b8:37:c2:3b:64:84:1f:60:
        1c:96:24:bf:07:fe:95:95:2a:33:33:cf:7d:eb:a1:b9:27:e1:
        9d:ba:8d:8b:5c:3c:9b:30:47:94:f5:f3:00:1b:8b:1c:28:95:
        08:3b:73:22:52:94:67:44:ea:39:1d:1b:1d:b5:b7:09:1c:34:
        6a:be:b9:d5:ec:88:95:98:c9:a6:7f:6d:a4:dc:2a:53:39:89:
        fb:bc:aa:81:1e:42:47:e4:53:60:a6:58:be:25:e0:c0:ea:16:
        1b:10:93:b0:c7:2a:b7:68:fc:3b:73:2d:f6:b5:f8:76:ec:8e:
        7b:fa:ef:39:b7:2e:b4:78:eb:05:e6:e6:c3:75:95:d0:6c:e1:
        7d:75:21:68:5f:97:01:0f:29:14:7f:d2:35:ef:a7:5d:f9:0d:
        c4:9e:f8:d6:e8:1f:ee:9a:70:6e:bf:53:73:67:2b:fb:9e:7a:
        0b:82:ba:d2:38:0b:81:a0:66:ba:5b:83:ad:91:aa:2b:b9:18:
        e0:ee:e1:05:77:cb:59:22:e1:0e:62:2e:93:30:35:53:94:c6:
        c3:84:1b:8a:33:59:87:13:0e:99:14:5a:9e:69:a4:6a:23:71:
        a5:65:88:f8:dc:13:53:c9:d9:61:3a:07:fe:ab:3d:f5:4d:64:
        ca:35:dd:39:98:0c:0a:b9:6d:78:5e:41:59:8d:76:d2:dc:88:
        e2:cd:7b:0b:26:e3:a0:c2:6f:a5:56:60:6b:e8:26:ac:28:7d:
        cb:d9:76:39:df:f9:d5:a5:28:38:3f:30:40:05:61:4d:01:1d:
        ed:53:0f:c4:fd:cf:f5:f7:8f:c3:94:31:29:cd:2f:cb:ea:f8:
        0c:39:89:81:2e:2c:32:16
```

12.2. CSRs and Certificates

In examples we assume that device has:

- Keys described in “12.1 Example keys used”.
- deviceId is 42.
- Assigned by RSL device name for use in CSR Subject CN is “RSL-SN0001-0000000042”.

12.2.1. ECC ECDSA on SECG secp256r1

12.2.1.1. CSR

ECC ECDSA on SECG secp256r1 CSR:


```
-----BEGIN CERTIFICATE REQUEST-----
MIHbMIGCAgEAMCAxHjAcBgNVBAMMFVJTTC1TTjAwMDEtMDAwMDAwMDA0MjBZMBMG
ByqGSM49AgEGCCqGSM49AwEHA0IABPu/cO9julGDiXPZnzvBIzBJfbsrCWlidJFX
pZSI+OmrZexQBuDaWFQ3ureaU0lpTOqgo+ainmpREimpTJ1d9qmgADAKBggqhkjO
PQQDAgNIADBFAiABO5zIgKsllL1i020M8ZKlnXQlKRlHikgE0sIL0ddWMwIhAOEo
z0qtLWLzzLNoTeKt9w4wza71neAVq8bZuO6x4gYh
-----END CERTIFICATE REQUEST-----
```


In textual representation form:

```
Certificate request self-signature verify OK
Certificate Request:
    Data:
        Version: 1 (0x0)
        Subject: CN=RSL-SN0001-0000000042
        Subject Public Key Info:
            Public Key Algorithm: id-ecPublicKey
                Public-Key: (256 bit)
                pub:
                    04:fb:bf:70:ef:63:ba:51:83:89:73:d9:9f:3b:c1:
                    23:30:49:7d:bb:2b:09:69:62:74:91:57:a5:94:88:
                    f8:e9:ab:65:ec:50:06:e0:da:58:54:37:ba:b7:9a:
                    53:49:69:4c:ea:a0:a3:e6:a2:9e:6a:51:12:29:a9:
                    4c:9d:5d:f6:a9
                ASN1 OID: prime256v1
                NIST CURVE: P-256
        Attributes:
            (none)
            Requested Extensions:
    Signature Algorithm: ecdsa-with-SHA256
    Signature Value:
        30:45:02:20:01:3b:9c:c8:80:ab:25:94:bd:62:d3:6d:0c:f1:
        92:a5:9d:74:25:29:19:47:8a:48:04:d2:c2:0b:d1:d7:56:33:
        02:21:00:e1:28:cf:4a:ad:2d:62:f3:cc:b3:68:4d:e2:ad:f7:
        0e:30:cd:ae:f5:9d:e0:15:ab:c6:d9:b8:ee:b1:e2:06:21
```

12.2.1.2. Certificate

ECC ECDSA on SECG secp256r1 Certificate:


```
-----BEGIN CERTIFICATE-----
MIIEFjCCAf6gAwIBAgIIJASrEs1sAAEwDQYJKoZIhvcNAQELBQAwUTELMAkGA1UE
BhMCTFMxITAfBgNVBAoMGFJldmVudWUgU2VydmljZXMgTGVzb3RobzEfMB0GA1UE
AwwWUlNMICoqU0FNUExFKiogUm9vdCBDQTAeFw0yNDA0MDIxNDQxMjZaFw0zNDA3
MDkxNDQxMjZaMFAxCzAJBgNVBAYTAkxTMSEwHwYDVQQKDBhSZXZlbnVlIFNlcnZp
Y2VzIExlc290aG8xHjAcBgNVBAMMFVJTTC1TTjAwMDEtMDAwMDAwMDA0MjBZMBMG
ByqGSM49AgEGCCqGSM49AwEHA0IABPu/cO9julGDiXPZnzvBIzBJfbsrCWlidJFX
pZSI+OmrZexQBuDaWFQ3ureaU0lpTOqgo+ainmpREimpTJ1d9qmjgb0wgbowHQYD
VR0OBBYEFGqv3pesJRu7AJBrDE7T/+dM4Kg1MB8GA1UdIwQYMBaAFI0g81HfZAuq
/6xhU6dOucqHFIOgMAwGA1UdEwEB/wQCMAAwDgYDVR0PAQH/BAQDAgXgMBMGA1Ud
JQQMMAoGCCsGAQUFBwMCMEUGCWCGSAGG+EIBDQQ4FjZPcGVuU1NMIEdlbmVyYXRl
ZCAqKlNBTVBMRSoqIEdXIEFQSSBDbGllbnQgQ2VydGlmaWNhdGUwDQYJKoZIhvcN
AQELBQADggIBAJyzB86kYYM13XV+m/rar4kALDWnN/C2DjFRakiDPFZDTtP28ynd
JALYsj6MJTQ6Y2Yy25DCH2MYECfranghyMm+3Zw9yG8EzaZtvwvEYi2NO9/pWmj6
bi6Q3CwItvElDVPro4rsm0yIyra71gyw9rxGpkftwOFliXPsAKCGkRH5qgwQBJ0c
D7Rnvug3pyyV9gSD+Qn9dbziuPDVjm1co8O1+UXgnw9wFR9ojCW/iwLwtrSWqWZs
TMAAuYENBJtanovnBgKAYP6XUyhPhVxGvjFZnlO4TC43W1MvXj6b34PnJs4TS1X5
edOol9qkELprqW+bi2DQ55nXSjIrQ0Ohrv6OasqnI1zy56s1MwpV5JTi5xDsSe+m
```

GZGOGpsl0gZIiHVUILErZz6Ktr52zwjmR8/Jxhy+EXhRIXTqyvqWwjqzo/aEHKC1
9Hdm4lORc4fTxx9HyAoGE/H0H6c0l0tiPbPAurdvVa1mIkuzO6DCOHJsmC/eXOep
NValHoHLIrBBzZVW5l1nsfjGmo9TelRQhTB6Esj80wYF6I94o116k2jAD7VhF2i+
ii4QejPHC4W9DGc5j9yZ8OF7l1nexH32rgFpFlylIGf27RUdaaIEdexDFyjy0p99
ODmXShgWpw2KqMumOZpFuv5ekvKKqx7/CfbTsgDLUzTRh0QhExpJJtTk
-----END CERTIFICATE-----

In textual representation form:

```
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            24:04:ab:12:cd:6c:00:01
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=LS, O=Revenue Services Lesotho, CN=RSL **SAMPLE** Root CA
        Validity
            Not Before: Apr  2 14:41:26 2024 GMT
            Not After : Jul  9 14:41:26 2034 GMT
        Subject: C=LS, O=Revenue Services Lesotho, CN=RSL-SN0001-0000000042
        Subject Public Key Info:
            Public Key Algorithm: id-ecPublicKey
                Public-Key: (256 bit)
                pub:
                    04:fb:bf:70:ef:63:ba:51:83:89:73:d9:9f:3b:c1:
                    23:30:49:7d:bb:2b:09:69:62:74:91:57:a5:94:88:
                    f8:e9:ab:65:ec:50:06:e0:da:58:54:37:ba:b7:9a:
                    53:49:69:4c:ea:a0:a3:e6:a2:9e:6a:51:12:29:a9:
                    4c:9d:5d:f6:a9
                ASN1 OID: prime256v1
                NIST CURVE: P-256
        X509v3 extensions:
            X509v3 Subject Key Identifier:
                6A:AF:DE:97:AC:25:1B:BB:00:90:6B:0C:4E:D3:FF:E7:4C:E0:A8:35
            X509v3 Authority Key Identifier:
                8D:20:F3:51:DF:64:0B:AA:FF:AC:61:53:A7:4E:B9:CA:87:14:83:A0
            X509v3 Basic Constraints: critical
                CA:FALSE
            X509v3 Key Usage: critical
                Digital Signature, Non Repudiation, Key Encipherment
            X509v3 Extended Key Usage:
                TLS Web Client Authentication
            Netscape Comment:
                OpenSSL Generated **SAMPLE** GW API Client Certificate
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:
        9c:b3:07:ce:a4:61:83:35:dd:75:7e:9b:fa:da:af:89:00:2c:
        35:a7:37:f0:b6:0e:31:51:6a:48:83:3c:56:43:4e:d3:f6:f3:
        29:dd:24:02:d8:b2:3e:8c:25:34:3a:63:66:32:db:90:c2:1f:
        63:18:10:27:eb:6a:78:21:c8:c9:be:dd:9c:3d:c8:6f:04:cd:
        a6:6d:bf:0b:c4:62:2d:8d:3b:df:e9:5a:68:fa:6e:2e:90:dc:
        2c:08:b6:f1:25:0d:53:eb:a3:8a:ec:9b:4c:88:ca:b6:bb:d6:
        0c:b0:f6:bc:46:a6:47:ed:c0:e1:65:89:73:ec:00:a0:86:91:
        11:f9:aa:0c:10:04:9d:1c:0f:b4:67:be:e8:37:a7:2c:95:f6:
        04:83:f9:09:fd:75:bc:e2:b8:f0:d5:8e:6d:5c:a3:c3:b5:f9:
        45:e0:9f:0f:70:15:1f:68:8c:25:bf:8b:02:f0:b6:b4:96:a9:
        66:6c:4c:c0:00:b9:81:0d:04:9b:5a:9e:8b:e7:06:02:80:60:
        fe:97:53:28:4f:85:5c:46:be:31:59:9e:53:b8:4c:2e:37:5b:
        53:2f:5e:3e:9b:df:83:e7:26:ce:13:4b:55:f9:79:d3:a8:97:
        da:a4:10:ba:6b:a9:6f:9b:8b:60:d0:e7:99:d7:4a:32:2b:43:
        43:a1:ae:fe:8e:6a:ca:a7:97:5c:f2:e7:ab:35:33:0a:55:e4:
```

```
                94:e2:e7:10:ec:49:ef:a6:19:91:8e:1a:9b:25:d2:06:48:88:
                75:54:20:b1:2b:67:3e:8a:b6:be:76:cf:08:e6:47:cf:c9:c6:
                1c:be:11:78:51:21:74:ea:ca:fa:96:c2:3a:b3:a3:f6:84:1c:
                a0:b5:f4:77:66:e2:53:91:73:87:d3:c7:1f:47:c8:0a:06:13:
                f1:f4:1f:a7:34:97:4b:62:3d:b3:c0:ba:b7:6f:55:ad:66:22:
                4b:b3:3b:a0:c2:38:72:6c:98:2f:de:5c:e7:a9:35:56:a5:1e:
                81:cb:22:b0:41:cd:95:56:e6:5d:67:b1:f8:c6:9a:8f:53:7a:
                54:50:85:30:7a:12:c8:fc:d3:06:05:e8:8f:78:a3:5d:7a:93:
                68:c0:0f:b5:61:17:68:be:8a:2e:10:7a:33:c7:0b:85:bd:0c:
                67:39:8f:dc:99:f0:e1:7b:97:59:de:c4:7d:f6:ae:01:69:16:
                5c:a5:20:67:f6:ed:15:1d:69:a2:04:75:ec:43:17:28:f2:d2:
                9f:7d:38:39:97:4a:18:16:a7:0d:8a:a8:cb:a6:39:9a:45:ba:
                fe:5e:92:f2:8a:ab:1e:ff:09:f6:d3:b2:00:cb:53:34:d1:87:
                44:21:13:1a:49:26:d4:e4
```

12.2.2. RSA 2048

12.2.2.1. CSR

RSA 2048 CSR:


```
-----BEGIN CERTIFICATE REQUEST-----
MIICZTCCAU0CAQAwIDEeMBwGA1UEAwwVUlNMLVNOMDAwMS0wMDAwMDAwMDQyMIIB
IjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEA52kCd2bXK+W72vC0+KlQ+tLU
BNBsYNk9Gg+NPTxfD+92lFg9sPW8B1MFxTO+Kpw5MRzArB6M3LZ3pj0O525vLcgT
301bridwTgpSfzqtoHFhgToxMy94lMiYSK94w2RxslaMyDm4dCXGfU5AlAiuuegV
Fz056jV/Ik7jFjQLG5GQnRhWtGkI2TKZLOlBsjhxUKLKPwUCPtkfZmDig/fXE1Xi
gqSOGMoxQ9BTxJ/i8wWU3AR1Tjou41EFisaOKpan4xqeRnvNrOs0eIccBFRXBCB7
8tLTZgE9yqgwoQ5oQpNm845OkFXQ0F770cy0sx+CO2OnpdQisRK8WDx95kJkIQID
AQABoAAwDQYJKoZIhvcNAQELBQADggEBAB+DHph6NeNh3/rE0YV+i8libG86CnlZ
gr+3Bknhn7uY9O0BdvoIzAq/403n1eD/DiWkciEW5ek8+aL8OTBid0HncgE9kNsz
OCR8FmBS4ZfmdMojgw4Fj7fV3EgoeOpdLWiUWs5SxKx5kYFK5k2Xql55vrkITeow
lurGRg3fguo3j9oNNIBMzcp9602CyCJxYCusfHP0/qaSljBv1/tx3Q4YvfLIqpHg
OV211bC+LgSAy4Oi5kUqI43TD3SguZhHfuOVbVwxAnSnPWVVmlWhebB/2LGKlF7O
IeknNW3HQb70KNRrFS8REkX88gAGCZW8zpGDRTn00eZW1lKHWd5eIL8=
-----END CERTIFICATE REQUEST-----
```


In textual representation form:

```
Certificate request self-signature verify OK
Certificate Request:
    Data:
        Version: 1 (0x0)
        Subject: CN=RSL-SN0001-0000000042
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (2048 bit)
                Modulus:
                    00:e7:69:02:77:66:d7:2b:e5:bb:da:f0:b4:f8:a9:
                    50:fa:d2:d4:04:d0:6c:60:d9:3d:1a:0f:8d:3d:3c:
                    5f:0f:ef:76:94:58:3d:b0:f5:bc:07:53:05:c5:33:
                    be:2a:9c:39:31:1c:c0:ac:1e:8c:dc:b6:77:a6:3d:
                    0e:e7:6e:6f:2d:c8:13:df:4d:5b:ae:27:70:4e:0a:
                    52:7f:3a:ad:a0:71:61:81:3a:31:33:2f:78:94:c8:
                    98:48:af:78:c3:64:71:b2:56:8c:c8:39:b8:74:25:
                    c6:7d:4e:40:94:08:ae:b9:e8:15:17:3d:39:ea:35:
                    7f:22:4e:e3:16:34:0b:1b:91:90:9d:18:56:b4:69:
                    08:d9:32:99:2c:e9:41:b2:38:71:50:a2:ca:3f:05:
                    02:3e:d9:1f:66:60:e2:83:f7:d7:13:55:e2:82:a4:
                    8e:18:ca:31:43:d0:53:c4:9f:e2:f3:05:94:dc:04:
                    75:4e:3a:2e:e3:51:05:8a:c6:8e:2a:96:a7:e3:1a:
                    9e:46:7b:cd:ac:eb:34:78:87:1c:04:54:57:04:20:
```

```
                           7b:f2:d2:d3:66:01:3d:ca:a8:30:a1:0e:68:42:93:
                           66:f3:8e:4e:90:55:d0:d0:5e:fb:d1:cc:b4:b3:1f:
                           82:3b:63:a7:a5:d4:22:b1:12:bc:58:3c:7d:e6:42:
                           64:21
                       Exponent: 65537 (0x10001)
               Attributes:
                   (none)
                   Requested Extensions:
           Signature Algorithm: sha256WithRSAEncryption
           Signature Value:
               1f:83:1e:98:7a:35:e3:61:df:fa:c4:d1:85:7e:8b:c9:62:6c:
               6f:3a:0a:79:59:82:bf:b7:06:49:e1:9f:bb:98:f4:ed:01:76:
               fa:08:cc:0a:bf:e3:4d:e7:d5:e0:ff:0e:25:a4:72:21:16:e5:
               e9:3c:f9:a2:fc:39:30:62:77:41:e7:72:01:3d:90:db:33:38:
               24:7c:16:60:52:e1:97:e6:74:ca:23:83:0e:05:8f:b7:d5:dc:
               48:28:78:ea:5d:2d:68:94:5a:ce:52:c4:ac:79:91:81:4a:e6:
               4d:97:aa:5e:79:be:b9:08:4d:ea:30:96:ea:c6:46:0d:df:82:
               ea:37:8f:da:0d:34:80:4c:cd:ca:7d:eb:4d:82:c8:22:71:60:
               2b:ac:7c:73:f4:fe:a6:92:96:30:6f:d7:fb:71:dd:0e:18:bd:
               f2:c8:aa:91:e0:39:5d:b5:d5:b0:be:2e:04:80:cb:83:a2:e6:
               45:2a:23:8d:d3:0f:74:a0:b9:98:47:7e:e3:95:6d:5c:31:02:
               74:a7:3d:65:55:9a:55:a1:79:b0:7f:d8:b1:8a:94:5e:ce:21:
               e9:27:35:6d:c7:41:be:f4:28:d4:6b:15:2f:11:12:45:fc:f2:
               00:06:09:95:bc:ce:91:83:45:39:f4:d1:e6:56:d6:52:87:59:
               de:5e:20:bf
```

12.2.2.2. Certificate

RSA 2048 Certificate:


```
-----BEGIN CERTIFICATE-----
MIIE4TCCAsmgAwIBAgIIJASrEs1sAAIwDQYJKoZIhvcNAQELBQAwUTELMAkGA1UE
BhMCTFMxITAfBgNVBAoMGFJldmVudWUgU2VydmljZXMgTGVzb3RobzEfMB0GA1UE
AwwWUUlNMICoqU0FNUExFKiogUm9vdCBDQTAeFw0yNDA0MDIxNTA2MzdaFw0zNDA3
MDkxNTA2MzdaMFAxCzAJBgNVBAYTAkxTMSEwHwYDVQQKDBhSZXZlbnVlIFNlcnZp
Y2VzIExlc290aG8xHjAcBgNVBAMMFVJTTC1TTjAwMDEtMDAwMDAwMDA0MjCCASIw
DQYJKoZIhvcNAQEBBQADggEPADCCAQoCggEBAOdpAndm1yvlu9rwtPipUPrS1ATQ
bGDZPRoPjT08Xw/vdpRYPbD1vAdTBcUzviqcOTEcwKwejNy2d6Y9Duduby3IE99N
W64ncE4KUn86raBxYYE6MTMveJTImEiveMNkcbJWjMg5uHQlxn1OQJQIrrnoFRc9
Oeo1fyJO4xY0CxuRkJ0YVrRpCNkymSzpQbI4cVCiyj8FAj7ZH2Zg4oP31xNV4oKk
jhjKMUPQU8Sf4vMFlNwEdU46LuNRBYrGjiqWp+MankZ7zazrNHiHHARUVwQge/LS
02YBPcqoMKEOaEKTZvOOTpBV0NBe+9HMtLMfgjtjp6XUIrESvFg8feZCZCECAwEA
AaOBvTCBujAdBgNVHQ4EFgQUMsDu2Cjv8kvPlrIVrgc/drsrJ/YwHwYDVR0jBBgw
FoAUjSDzUd9kC6r/rGFTp065yocUg6AwDAYDVR0TAQH/BAIwADAOBgNVHQ8BAf8E
BAMCBeAwEwYDVR0lBAwwCgYIKwYBBQUHAwIwRQYJYIZIAYb4QgENBDgWNk9wZW5T
U0wgR2VuZXJhdGVkICoqU0FNUExFKiogR1cgQVBJIENsaWVudCBDZXJ0aWZpY2F0
ZTANBgkqhkiG9w0BAQsFAAOCAgEAIv6zwNN8X2Nfd8+HTaeFgQlxLS8AobnuM8Zm
9tI30v2oie8HSK6kpNbvWDj/asfwwn2xp+jogvW8JqV8hPwskVSC/rLgeRzZZnGY
9iNRxjrEivSxlYp2I8ZwtbSOq69JpcX0ekSg1XhhRVIUIt0HWHk7eFcwUVn9Z6fl
F4LDQYqLrEGO9+vhxrh7XnuBR2o0GACODRdeqlL+IpwScEaB7/3Bh3QVqyjjnqO9
WfXqvNn2JO1bj95FGvO8n8Rn7iOBRcbioPOjS9AyDAjwdkgCleEIuRutYn0t9ncw
Y1e5ggnkOjwIv/tlX5MO6M+2EqJCtPmNcFCR1N2P1lfKZxotvXLd21COIp2PU5p/
sZNZNPNW7dqKO6sUvdvwoH2UkdSONKE5ccsz/C6+XcH7eZvUcFIzUCuIBnk7xGTT
LPU3syv6ikWYxrxAP9oE8wU0x0kdGnjsReccZqOYH2a5RDTk6lLbbkBGtUn0RjJ2
zMMNm7ZKdglon6mJbHtIKgaxvq3DyDA/Z12I/rMh3jjFuQARSuJ2QVc3+Y0yLt4w
BaQ88QnkkywqivcAOkWXijYC+vTMQvOtxs1STcNxd5b2z37LS5betFeSzRjBz6Wd
DxhWfeAHiRsVeBPJtCpR1AR1MQ1U5YkHLayskP7NIHPc0yzrEyJgCWgOUBPSvrGQ
WWABhow=
-----END CERTIFICATE-----
```


In textual representation form:

```
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            24:04:ab:12:cd:6c:00:02
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=LS, O=Revenue Services Lesotho, CN=RSL **SAMPLE** Root CA
        Validity
            Not Before: Apr  2 15:06:37 2024 GMT
            Not After : Jul  9 15:06:37 2034 GMT
        Subject: C=LS, O=Revenue Services Lesotho, CN=RSL-SN0001-0000000042
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (2048 bit)
                Modulus:
                    00:e7:69:02:77:66:d7:2b:e5:bb:da:f0:b4:f8:a9:
                    50:fa:d2:d4:04:d0:6c:60:d9:3d:1a:0f:8d:3d:3c:
                    5f:0f:ef:76:94:58:3d:b0:f5:bc:07:53:05:c5:33:
                    be:2a:9c:39:31:1c:c0:ac:1e:8c:dc:b6:77:a6:3d:
                    0e:e7:6e:6f:2d:c8:13:df:4d:5b:ae:27:70:4e:0a:
                    52:7f:3a:ad:a0:71:61:81:3a:31:33:2f:78:94:c8:
                    98:48:af:78:c3:64:71:b2:56:8c:c8:39:b8:74:25:
                    c6:7d:4e:40:94:08:ae:b9:e8:15:17:3d:39:ea:35:
                    7f:22:4e:e3:16:34:0b:1b:91:90:9d:18:56:b4:69:
                    08:d9:32:99:2c:e9:41:b2:38:71:50:a2:ca:3f:05:
                    02:3e:d9:1f:66:60:e2:83:f7:d7:13:55:e2:82:a4:
                    8e:18:ca:31:43:d0:53:c4:9f:e2:f3:05:94:dc:04:
                    75:4e:3a:2e:e3:51:05:8a:c6:8e:2a:96:a7:e3:1a:
                    9e:46:7b:cd:ac:eb:34:78:87:1c:04:54:57:04:20:
                    7b:f2:d2:d3:66:01:3d:ca:a8:30:a1:0e:68:42:93:
                    66:f3:8e:4e:90:55:d0:d0:5e:fb:d1:cc:b4:b3:1f:
                    82:3b:63:a7:a5:d4:22:b1:12:bc:58:3c:7d:e6:42:
                    64:21
                Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Subject Key Identifier:
                32:C0:EE:D8:28:EF:F2:4B:CF:96:B2:15:AE:07:3F:76:BB:2B:27:F6
            X509v3 Authority Key Identifier:
                8D:20:F3:51:DF:64:0B:AA:FF:AC:61:53:A7:4E:B9:CA:87:14:83:A0
            X509v3 Basic Constraints: critical
                CA:FALSE
            X509v3 Key Usage: critical
                Digital Signature, Non Repudiation, Key Encipherment
            X509v3 Extended Key Usage:
                TLS Web Client Authentication
            Netscape Comment:
                OpenSSL Generated **SAMPLE** GW API Client Certificate
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:
        22:fe:b3:c0:d3:7c:5f:63:5f:77:cf:87:4d:a7:85:81:09:71:
        2d:2f:00:a1:b9:ee:33:c6:66:f6:d2:37:d2:fd:a8:89:ef:07:
        48:ae:a4:a4:d6:ef:58:38:ff:6a:c7:f0:c2:7d:b1:a7:e8:e8:
        82:f5:bc:26:a5:7c:84:fc:2c:91:54:82:fe:b2:e0:79:1c:d9:
        66:71:98:f6:23:51:c6:3a:c4:8a:f4:b1:95:8a:76:23:c6:70:
        b5:b4:8e:ab:af:49:a5:c5:f4:7a:44:a0:d5:78:61:45:52:14:
        22:dd:07:58:79:3b:78:57:30:51:59:fd:67:a7:e5:17:82:c3:
        41:8a:8b:ac:41:8e:f7:eb:e1:c6:b8:7b:5e:7b:81:47:6a:34:
        18:00:8e:0d:17:5e:aa:52:fe:22:9c:12:70:46:81:ef:fd:c1:
        87:74:15:ab:28:e3:9e:a3:bd:59:f5:ea:bc:d9:f6:24:ed:5b:
        8f:de:45:1a:f3:bc:9f:c4:67:ee:23:81:45:c6:e2:a0:f3:a3:
```

```
4b:d0:32:0c:08:f0:76:48:02:95:e1:08:b9:1b:ad:62:7d:2d:
f6:77:30:63:57:b9:82:09:e4:3a:3c:08:bf:fb:65:5f:93:0e:
e8:cf:b6:12:a2:42:b4:f9:8d:70:50:91:d4:dd:8f:d6:57:ca:
67:1a:2d:bd:72:dd:db:50:8e:22:9d:8f:53:9a:7f:b1:93:59:
34:f3:56:ed:da:8a:3b:ab:14:bd:db:f0:a0:7d:94:91:d4:8e:
34:a1:39:71:cb:33:fc:2e:be:5d:c1:fb:79:9b:d4:70:52:33:
50:2b:88:06:79:3b:c4:64:d3:2c:f5:37:b3:2b:fa:8a:45:98:
c6:bc:40:3f:da:04:f3:05:34:c7:49:1d:1a:78:ec:45:e7:1c:
66:a3:98:1f:66:b9:44:34:e4:ea:52:db:6e:40:46:b5:49:f4:
46:32:76:cc:c3:0d:9b:b6:4a:76:09:68:9f:a9:89:6c:7b:48:
2a:06:b1:be:ad:c3:c8:30:3f:67:5d:88:fe:b3:21:de:38:c5:
b9:00:11:4a:e2:76:41:57:37:f9:8d:32:2e:de:30:05:a4:3c:
f1:09:e4:93:2c:2a:8a:f7:00:3a:45:97:8a:36:02:fa:f4:cc:
42:f3:ad:c6:cd:52:4d:c3:71:77:96:f6:cf:7e:cb:4b:96:de:
b4:57:92:cd:18:c1:cf:a5:9d:0f:18:56:7d:e0:07:89:1b:15:
78:13:c9:b4:2a:51:d4:04:75:31:0d:54:e5:89:07:2d:ac:ac:
90:fe:cd:20:73:dc:d3:2c:eb:13:22:60:09:68:0e:50:13:d2:
be:b1:90:59:60:01:86:8c
```

13. SIGNATURES GENERATION AND VERIFICATION RULES

13.1. Signature and hash generation algorithm

Below algorithm is used to generate receipt and fiscal period hash and signature:

- Receipt or fiscal period fields must be converted to string (by rules as described in table below) and concatenated (no concatenation character is used);

1) Concatenated line must be hashed using SHA256;
2) Hash must be signed with private key.

Formula to get a hash: Hash = SHA-256(x_1|| x_2||…||x_n);
Formula to get a signature:
Signature = RSA(Hash,d,n) – in case RSA keys are used
or
Signature = ECC(Hash,CURVE,g,n) – in case ECC keys are used

Where
|| - means field concatenation;
x_1, x_2, …, x_n - receipt or fiscal period fields;
d – secret RSA exponent;
n - RSA modulus
CURVE – the elliptic curve field and equation used
G – elliptic curve base point, a point on the curve that generates a subgroup of large prime order n
n – integer order of G, means that n x G=O, where O is the identity element.

13.2. Receipt signature generation and verification

Receipt hash and signature are generated according to the rules provided in section 13.1.

13.2.1. Receipt device signature

Fields included in receipt hash which is used for device signature are (these fields must be included in hash in the same order as provided below):

Order	Field name	Description
1	deviceID	Device ID
2	receiptType	Receipt type value in upper case.
3	receiptCurrency	Currency code (ISO 4217 currency code). Must be LSL. It must be in upper case.
4	receiptGlobalNo	Receipt global number.
5	receiptDate	Date in ISO 8601 format <date>T<time>, YYYY-MM-DDTHH:mm:ss (hours are represented in 24 hours format, local time). Example: 2024-02-23T14:43:23
6	receiptTotal	Receipt total is included in signature in cents. Examples: - If receiptTotal is 500 LSL, value 50000 must be used in signature. - If receiptTotal is 12,34 LSL, value 1234 must be used in signature. - If receiptTotal is 0,05 LSL, value 5 must be used in signature.
7	receiptTaxes	Concatenated receiptTaxes, where each line is concatenated in this way: taxCode || taxRate || taxAmount || salesAmountWithTax. In case of taxRate is not sent, empty value should be used in signature. Amounts are represented in cents. In case taxRate is not an integer there should be dot between the integer and fractional part. In case of exempt

		which does not send tax percent value, empty value should be used in signature. In case taxRate is an integer there should be value of tax percent, dot and two zeros sent. Taxes are ordered by taxID in ascending order and taxCode in alphabetical order (if taxCode is empty it is ordered before A letter). Examples: - If taxRate is 15, value 15.00 must be used in signature. - If taxRate is 14.5 value 14.50 must be used in signature. - If taxRate is 0 value 0.00 must be used in signature.
8	previousReceiptHash	Previous receipt hash is included into current receipt device signature. This will create a chain of receipts. This field is not used in signature when current receipt is first in fiscal period.

FiscalInvoice Examples:

Name	Example No 1
deviceID	321
receiptType	FISCALINVOICE
receiptCurrency	LSL
receiptGlobalNo	432
receiptDate	2024-02-28T15:43:12
receiptTotal	9450,00
receiptTaxes	Tax lines:

taxID	taxCode	taxRate	taxAmount	salesAmountWithTax
1	A		0,00	2500,00
2	B	0	0,00	3500,00
3	C	15	150,00	1150,00
3	D	15	300,00	2300,00

	Result: A0250000B0.000350000C15.0015000115000D15.0030000230000
previousReceiptHash	hNVJXP/ACOiE8McD3pKsDlqBXpuaUqQOfPnMyfZWI9k=
Result used for hash generation	321FISCALINVOICELSL4322024-02-28T15:43:12945000A0250000B0.000350000C15.0015000115000D15.0030000230000 hNVJXP/ACOiE8McD3pKsDlqBXpuaUqQOfPnMyfZWI9k=
Generated receipt hash in base64 representation	hFFu8/qwBh9yxPKSnkDVUG1HSmzCpHJfGEwqXCiAdK0=

Name	Example No 2
deviceID	322
receiptType	FISCALINVOICE
receiptCurrency	LSL
receiptGlobalNo	85
receiptDate	2024-02-19T09:23:07
receiptTotal	40,35
receiptTaxes	Tax lines:

taxID	taxRate	taxAmount	salesAmountWithTax
1		0,00	7,00
2	0	0,00	10,00
3	14,5	0,05	0,35

	Result: 07000.000100014.50535
previousReceiptHash	hNVJXP/ACOiE8McD3pKsDlqBXpuaUqQOfPnMyfZWI9k=
Result used for hash generation	322FISCALINVOICELSL852024-02-19T09:23:07403507000.000100014.50535 hNVJXP/ACOiE8McD3pKsDlqBXpuaUqQOfPnMyfZWI9k=
Generated receipt hash in base64 representation	KTDotht93aZditmoe+TWzbRlD/orX6d0Pu/aHkwpAy0=

Name	Example No 3
deviceID	322
receiptType	FISCALINVOICE
receiptCurrency	LSL
receiptGlobalNo	85
receiptDate	2024-02-19T09:23:07
receiptTotal	50,35
receiptTaxes	Tax lines: <table><tr><td>taxID</td><td>taxRate</td><td>taxAmount</td><td>salesAmountWithTax</td></tr><tr><td>1</td><td></td><td>0,00</td><td>7,00</td></tr><tr><td>2</td><td>0</td><td>0,00</td><td>10,00</td></tr><tr><td>3</td><td>14,5</td><td>0,05</td><td>0,35</td></tr><tr><td>4</td><td>10</td><td>10,00</td><td>10,00</td></tr></table> Result: 07000.000100014.5053510.0010001000
previousReceiptHash	hNVJXP/ACOiE8McD3pKsDlqBXpuaUqQOfPnMyfZWI9k=
Result used for hash generation	322FISCALINVOICELSL852024-02-19T09:23:07503507000.000100014.5053510.0010001000hNVJXP/ACOiE8McD3pKsDlqBXpuaUqQOfPnMyfZWI9k=
Generated receipt hash in base64 representation	1ExxQU5/mXJiM40sWAwxdBNkSqIj2qCjAWJXt0+V4g4=

CreditNote Examples:

Name	Example No 1
deviceID	321
receiptType	CREDITNOTE
receiptCurrency	LSL
receiptGlobalNo	432
receiptDate	2024-02-19T15:43:12
receiptTotal	-9450,00
receiptTaxes	Tax lines: <table><tr><td>taxID</td><td>taxCode</td><td>taxRate</td><td>taxAmount</td><td>salesAmountWithTax</td></tr><tr><td>1</td><td>A</td><td></td><td>0,00</td><td>-2500,00</td></tr><tr><td>2</td><td>B</td><td>0</td><td>0,00</td><td>-3500,00</td></tr><tr><td>3</td><td>C</td><td>15</td><td>-150,00</td><td>-1150,00</td></tr><tr><td>3</td><td>D</td><td>15</td><td>-300,00</td><td>-2300,00</td></tr></table>Result: A0-250000B0.000-350000C15.00-15000-115000D15.00-30000-230000
previousReceiptHash	hNVJXP/ACOiE8McD3pKsDlqBXpuaUqQOfPnMyfZWI9k=
Result used for hash generation	321CREDITNOTELSL4322024-02-19T15:43:12-945000 A0-250000B0.000-350000C15.00-15000-115000D15.00-30000-230000hNVJXP/ACOiE8McD3pKsDlqBXpuaUqQOfPnMyfZWI9k=
Generated receipt hash in base64 representation	4CWvbAGPwbVdbFTPLVk1UhaP+4KE2zCDafaD1+qhyVY=

Name	Example No 2
deviceID	322
receiptType	CREDITNOTE
receiptCurrency	LSL
receiptGlobalNo	85
receiptDate	2024-02-19T09:23:07
receiptTotal	-40,35
receiptTaxes	Tax lines: <table><tr><td>taxID</td><td>taxRate</td><td>taxAmount</td><td>salesAmountWithTax</td></tr><tr><td>1</td><td></td><td>0,00</td><td>-7,00</td></tr><tr><td>2</td><td>0</td><td>0,00</td><td>-10,00</td></tr></table>

<table>
<tr><td></td><td><table><tr><td>3</td><td>14,5</td><td>-3,00</td><td>-23,00</td></tr></table>Result:
0-7000.000-100014.50-300-2300</td></tr>
<tr><td>previousReceiptHash</td><td>hNVJXP/ACOiE8McD3pKsDlqBXpuaUqQOfPnMyfZWI9k=</td></tr>
<tr><td>Result used for hash generation</td><td>322CREDITNOTELSL852024-02-19T09:23:07-40350-7000.000-100014.50-300-2300hNVJXP/ACOiE8McD3pKsDlqBXpuaUqQOfPnMyfZWI9k=</td></tr>
<tr><td>Generated receipt hash in base64 representation</td><td>N4dMwVTbSXCr7qYPGj3sgLNkBKQDt1YlPnoz8xz8fYI=</td></tr>
</table>

DebitNote Examples:

<table>
<tr><th>Name</th><th>Example No 1</th></tr>
<tr><td>deviceID</td><td>321</td></tr>
<tr><td>receiptType</td><td>DEBITNOTE</td></tr>
<tr><td>receiptCurrency</td><td>LSL</td></tr>
<tr><td>receiptGlobalNo</td><td>432</td></tr>
<tr><td>receiptDate</td><td>2024-02-19T15:43:12</td></tr>
<tr><td>receiptTotal</td><td>9450,00</td></tr>
<tr><td>receiptTaxes</td><td>Tax lines:<table><tr><td>taxID</td><td>taxCode</td><td>taxRate</td><td>taxAmount</td><td>salesAmountWithTax</td></tr><tr><td>1</td><td>A</td><td></td><td>0,00</td><td>2500,00</td></tr><tr><td>2</td><td>B</td><td>0</td><td>0,00</td><td>3500,00</td></tr><tr><td>3</td><td>C</td><td>15</td><td>150,00</td><td>1150,00</td></tr><tr><td>3</td><td>C</td><td>15</td><td>300,00</td><td>2300,00</td></tr></table>Result:
A0250000B0.000350000C15.0015000115000D15.0030000230000</td></tr>
<tr><td>previousReceiptHash</td><td>hNVJXP/ACOiE8McD3pKsDlqBXpuaUqQOfPnMyfZWI9k=</td></tr>
<tr><td>Result used for hash generation</td><td>321DEBITNOTELSL4322024-02-19T15:43:12945000
A0250000B0.000350000C15.0015000115000D15.0030000230000hNVJXP/ACOiE8McD3pKsDlqBXpuaUqQOfPnMyfZWI9k=</td></tr>
<tr><td>Generated receipt hash in base64 representation</td><td>OC9auWXDqitEncmdwN9MXWAiQp8plsH4hC8fAeLLd3k=</td></tr>
</table>

<table>
<tr><th>Name</th><th>Example No 2</th></tr>
<tr><td>deviceID</td><td>322</td></tr>
<tr><td>receiptType</td><td>DEBITNOTE</td></tr>
<tr><td>receiptCurrency</td><td>LSL</td></tr>
<tr><td>receiptGlobalNo</td><td>85</td></tr>
<tr><td>receiptDate</td><td>2024-02-19T09:23:07</td></tr>
<tr><td>receiptTotal</td><td>40,35</td></tr>
<tr><td>receiptTaxes</td><td>Tax lines:<table><tr><td>taxID</td><td>taxRate</td><td>taxAmount</td><td>salesAmountWithTax</td></tr><tr><td>1</td><td></td><td>0,00</td><td>7,00</td></tr><tr><td>2</td><td>0</td><td>0,00</td><td>10,00</td></tr><tr><td>3</td><td>14,5</td><td>3,00</td><td>23,00</td></tr></table>Result:
07000.000100014.503002300</td></tr>
<tr><td>previousReceiptHash</td><td>hNVJXP/ACOiE8McD3pKsDlqBXpuaUqQOfPnMyfZWI9k=</td></tr>
<tr><td>Result used for hash generation</td><td>322DEBITNOTELSL852024-02-19T09:23:
07403507000.000100014.503002300hNVJXP/ACOiE8McD3pKsDlqBXpuaUqQOfPnMyfZWI9k=</td></tr>
<tr><td>Generated receipt hash in base64 representation</td><td>78mzcw/g3khBhrch8eiPPaK46aXe/hLXjbQyiT63eM8=</td></tr>
</table>

13.2.2. Receipt Lekuka signature

Receipt Lekuka signature may be verified by decrypting receiptServerSignature with Lekuka public key and comparing if it matches with prepared receipt hash.

receiptServerSignature is generated only for receipt submitted in “Online” receipt mode. Hash generation algorithm is provided in section 13.1.

Fields included in receipt hash which is used for Lekuka signature are (these fields must be included in hash in the same order as provided below):

Order	Field name	Description
1	receiptDeviceSignature	Receipt signature generated by device.
2	receiptID	Receipt ID
3	serverDate	Date in ISO 8601 format <date>T<time>, YYYY-MM-DDThh:mm:ss (hours are represented in 24 hours format, local time). Example: 2024-02-23T14:43:23

Example

Name	Example
receiptDeviceSignature	YyXTSizBBrMjMk4VQL+sCNr+2AC6aQbDAn9JMV2rk3yJ6MDZwie0wqQW3oisNWrMkeZsuAyFSnFkU2A+pKm91sOHVdjeRBebjQgAQQIMTCVIcYrx+BizQ7Ib9iCdsVI+Jel2nThqQiQzfRef6EgtgsaIAN+PV55xSrHvPkIe+Bc=
receiptID	48377
serverDate	2024-02-19T15:43:12
Result used for hash generation	YyXTSizBBrMjMk4VQL+sCNr+2AC6aQbDAn9JMV2rk3yJ6MDZwie0wqQW3oisNWrMkeZsuAyFSnFkU2A+pKm91sOHVdjeRBebjQgAQQIMTCVIcYrx+BizQ7Ib9iCdsVI+Jel2nThqQiQzfRef6EgtgsaIAN+PV55xSrHvPkIe+Bc=483772024-02-19T15:43:12
Generated hash in base64 representation	G8ZCt/U2pVW93/fd8Uyb4Gs9qm7zYL3YqDHkxiHndO4=

13.3. Fiscal period signature generation and verification

Fiscal period report hash and signature are generated according to the rules provided in section 13.1.

13.3.1. Fiscal period device signature

Fields included in fiscal period hash used for device signature are provided below (these fields must be included in hash in the same order as provided below):

Order	Field name	Description
1	deviceID	Device ID
2	fiscalDayNo	Fiscal period number
3	fiscalDayDate	Fiscal period date (date when fiscal period was opened). Date in ISO 8601 format YYYY-MM-DD. Example: 2024-02-23
4	fiscalDayCounters	Concatenated fiscal period counter lines, where each line is concatenated in this way: fiscalCounterType || fiscalCounterCurrency || fiscalCounterTaxRate or fiscalCounterMoneyType || fiscalCounterValue. All text values are concatenated in upper case. Amounts are represented in cents. Only non-zero value counters are included in concatenation. Counters are concatenated in this order: • fiscalCounterType (in ascending order, by enum digital value) • fiscalCounterCurrency (in alphabetical ascending order) • fiscalCounterTaxID (in ascending order) / fiscalCounterMoneyType (in ascending order, by enum digital value) In case taxRate is not an integer there should be dot between the integer and fractional part. In case of exempt which does not send tax percent value, empty value should be used in signature. In case taxRate is an integer there should be value of tax percent, dot and two zeros sent. Examples:

		- If taxRate is 15, value 15.00 must be used in signature. - If taxRate is 14.5 value 14.50 must be used in signature. - If taxRate is 0 value 0.00 must be used in signature.

Example:

<table>
<tr><th>Name</th><th>Example</th></tr>
<tr><td>deviceID</td><td>321</td></tr>
<tr><td>fiscalDayNo</td><td>84</td></tr>
<tr><td>fiscalDayDate</td><td>2024-02-23</td></tr>
<tr><td>fiscalDayCounters</td><td>
<table>
<tr><td>fiscalCounterType</td><td>fiscalCounterCurrency</td><td>fiscalCounterTaxRate/ fiscalCounterMoneyType</td><td>fiscalCounterValue</td></tr>
<tr><td>SaleByTax</td><td>LSL</td><td></td><td>23000,00</td></tr>
<tr><td>SaleByTax</td><td>LSL</td><td>0</td><td>12000,00</td></tr>
<tr><td>SaleByTax</td><td>LSL</td><td>14,5</td><td>25,00</td></tr>
<tr><td>SaleByTax</td><td>LSL</td><td>15</td><td>12,00</td></tr>
<tr><td>SaleTaxByTax</td><td>LSL</td><td>14,5</td><td>2,50</td></tr>
<tr><td>SaleTaxByTax</td><td>LSL</td><td>15</td><td>2300,00</td></tr>
<tr><td>BalanceByMoneyType</td><td>LSL</td><td>CASH</td><td>20000,00</td></tr>
<tr><td>BalanceByMoneyType</td><td>LSL</td><td>CARD</td><td>15000,00</td></tr>
<tr><td>PayoutByTax</td><td>LSL</td><td>15</td><td>-100,00</td></tr>
<tr><td>PayoutTaxByTax</td><td>LSL</td><td>15</td><td>-15,00</td></tr>
</table>
Result:
SALEBYTAXLSL2300000SALEBYTAXLSL0.001200000SALEBYTAXLSL14.502500SALEBYTAXLSL15.001200SALETAXBYTAXLSL14.50250SALETAXBYTAXLSL15.00230000BALANCEBYMONEYTYPELSLCASH2000000BALANCEBYMONEYTYPELSLCARD1500000PAYOUTBYTAXLSL15-10000PAYOUTTAXBYTAXLSL15-1500</td></tr>
<tr><td>Result used for hash generation</td><td>321842024-02-23SALEBYTAXLSL2300000SALEBYTAXLSL0.001200000SALEBYTAXLSL14.502500SALEBYTAXLSL15.001200SALETAXBYTAXLSL14.50250SALETAXBYTAXLSL15.00230000BALANCEBYMONEYTYPELSLCASH2000000BALANCEBYMONEYTYPELSLCARD1500000PAYOUTBYTAXLSL15-10000PAYOUTTAXBYTAXLSL15-1500</td></tr>
<tr><td>Generated hash in base64 representation</td><td>66gBvoNQIEPmOTBgvTCTIMg/XdqRp163Eh93g3JalJs=</td></tr>
</table>

13.3.2. Fiscal period Lekuka signature

Fiscal period Lekuka signature may be verified by decrypting fiscalDayServerSignature with Lekuka public key and comparing if it matches with prepared fiscal period hash.

Hash generation algorithm is provided in section 13.1.

Fields included in fiscal period hash used for Lekuka signature are provided below (these fields must be included in hash in the same order as provided below):

Order	Field name	Description
1	deviceID	Device ID
2	fiscalDayNo	Fiscal period number
3	fiscalDayDate	Fiscal period date (date when fiscal period was opened). Date in ISO 8601 format YYYY-MM-DD. Example: 2024-02-23
4	fiscalDayUpdated	Date and time when fiscal period was closed. Date in ISO 8601 format <date>T<time>, YYYY-MM-DDThh:mm:ss (hours are represented in 24 hours format, local time). Example: 2024-02-23T14:43:23 fiscalDayClosed value from response to device.
5	reconciliationMode	Defines how fiscal period was closed: automatically or manually. Possible values (in upper case): - AUTO - MANUAL
6	fiscalDayCounters	Concatenated fiscal period counter lines as described above in device signature generation.
7	fiscalDayDeviceSignature	Fiscal period signature generated by device. In case fiscal period is closed manually, this field is not included into hash for Lekuka signature.

Example:

<table>
<tr><th>Name</th><th>Example</th></tr>
<tr><td>deviceID</td><td>321</td></tr>
<tr><td>fiscalDayNo</td><td>84</td></tr>
<tr><td>fiscalDayDate</td><td>2024-02-23</td></tr>
<tr><td>fiscalDayUpdated</td><td>2024-02-23T14:43:23</td></tr>
<tr><td>reconciliationMode</td><td>AUTO</td></tr>
<tr><td>fiscalDayCounters</td><td>
<table>
<tr><td>fiscalCounterType</td><td>fiscalCounterCurrency</td><td>fiscalCounterTaxRate/ fiscalCounterMoneyType</td><td>fiscalCounterValue</td></tr>
<tr><td>SaleByTax</td><td>LSL</td><td></td><td>23000,00</td></tr>
<tr><td>SaleByTax</td><td>LSL</td><td>0</td><td>12000,00</td></tr>
<tr><td>SaleByTax</td><td>LSL</td><td>15</td><td>12,00</td></tr>
<tr><td>SaleTaxByTax</td><td>LSL</td><td>15</td><td>2300,00</td></tr>
<tr><td>BalanceByMoneyType</td><td>LSL</td><td>CARD</td><td>15000,00</td></tr>
<tr><td>BalanceByMoneyType</td><td>LSL</td><td>CASH</td><td>20000,00</td></tr>
<tr><td>PayoutByTax</td><td>LSL</td><td>15</td><td>-100,00</td></tr>
<tr><td>PayoutTaxByTax</td><td>LSL</td><td>15</td><td>-15,00</td></tr>
</table>
Result:
SALEBYTAXLSL2300000SALEBYTAXLSL0.001200000SALEBYTAXLSL15.001200SALETAXBYTAXLSL15.00230000BALANCEBYMONEYTYPELSLCARD1500000BALANCEBYMONEYTYPELSLCASH2000000PAYOUTBYTAXLSL15-10000PAYOUTTAXBYTAXLSL15-1500</td></tr>
<tr><td>fiscalDayDeviceSignature</td><td>YyXTSizBBrMjMk4VQL+sCNr+2AC6aQbDAn9JMV2rk3yJ6MDZwie0wqQW3oisNWrMkeZsuAyFSnFkU2A+pKm91sOHVdjeRBebjQgAQQIMTCVIcYrx+BizQ7Ib9iCdsVI+Jel2nThqQiQzfRef6EgtgsaIAN+PV55xSrHvPkIe+Bc=</td></tr>
<tr><td>Result used for hash generation</td><td>321842024-02-232024-02-23T22:21:14AUTOSALEBYTAXLSL2300000SALEBYTAXLSL0.001200000SALEBYTAXLSL15.001200SALETAXBYTAXLSL15.00230000BALANCEBYMONEYTYPELSLCARD1500000BALANCEBYMONEYTYPELSLCASH2000000PAYOUTBYTAXLSL15-10000PAYOUTTAXBYTAXLSL15-1500YyXTSizBBrMjMk4VQL+sCNr+2AC6aQbDAn9JMV2rk3yJ6MDZwie0wqQW3oisNWrMkeZsuAyFSnFkU2A+pKm91sOHVdjeRBebjQgAQQIMTCVIcYrx+BizQ7Ib9iCdsVI+Jel2nThqQiQzfRef6EgtgsaIAN+PV55xSrHvPkIe+Bc=</td></tr>
<tr><td>Generated hash in base64 representation</td><td>1xmEiH4XdKU8c+SzniByufe23oHBRvvwxIAds7OYQlA=</td></tr>
</table>